

A recurrent vision transformer trained with reinforcement and temporal self-supervision exhibits the signatures of primate visual attention

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Abstract

How attention improves perception has been debated for a quarter century: proposals range from enhancement of the sensory signal, to reduction of external noise and more efficient selection, to shifts in the observer’s decision criterion and reductions of spatial uncertainty. These accounts are not mutually exclusive — all are thought to contribute — but behaviour alone cannot cleanly separate them. Here we show that an affine-feedback recurrent vision-transformer architecture with a spatial working-memory state, trained with reinforcement learning plus a temporal joint-embedding predictive auxiliary loss (weight 0.5), reproduces several signatures of primate visual attention across separately trained checkpoints. Because its internal attention can be read out and perturbed, the model directly operationalizes two components of attentional performance: its decision criterion shifts toward liberal reporting and its perceptual sensitivity (d') depends on the attention clamp. Injected attention also evokes false detections preferentially at the monitored location, demonstrating spatial gating. These measurements do not constitute a quantitative three-way decomposition and include no dedicated external-noise test. The model deploys attention to a cued location, uses spatial validity when validity is the operative cue, maintains a value code from cue onset, and develops a decodable, position-specific code for the change when the change occurs. In a preliminary cued analogue of a paradigm used in the primate literature, separately trained reward-regime checkpoints reproduce the Luo–Maunsell sensitivity/criterion contrast. This RViT+ study extends the 2025 empirical Recurrent ViT [1]; it is also a companion to, but not a revision of, a separate 2026 normative account of when value-directed attention matters [2].

1 Introduction

For a quarter of a century, a single question has organised the study of visual attention: *how* does attention improve perception? Attending a location makes an observer faster and more accurate at the events that occur there, and the size of that benefit has been mapped in exhaustive psychophysical and physiological detail [3, 4]. What the benefit *is*—the computation that turns a spatial cue into better vision—has proven far harder to pin down. Reviewing the field, Carrasco [3] frames three standing accounts. On the first, attention *enhances the signal*: it multiplies the gain of the neurons that encode the attended stimulus, sharpening their response the way a contrast increase would [5, 6, 7, 8]. On the second, attention improves sensitivity by *reducing external noise*, or equivalently by selecting the relevant input more efficiently and excluding distractors and competing locations [9]. On the third, the gain in performance is not perceptual at all but sits at the *decision stage*—a shift in the observer’s response criterion, or a reduction in spatial uncertainty about where the relevant event will occur [10, 4].

These are not idle alternatives. They make different predictions about which neurons should change, about how the psychometric and the criterion should move, and about where in the processing stream an intervention should have its effect. Yet Carrasco’s own resolution is appropriately plural: the three accounts, she writes, “are not mutually exclusive; rather, all three are

likely to contribute” [3]. Behaviour is a summary of a system with many internal stages, and a change in hit rate can be consistent with a gain change, a noise change, or a criterion change. Signal-detection analysis can, with care, separate a sensitivity (d') effect from a criterion (c) effect [10]; dedicated external-noise paradigms are needed to adjudicate signal enhancement against external-noise reduction. An instrumented model adds a different form of access: its internal representation can be read out and its attention can be injected or suppressed at a chosen location. That access supports direct tests of criterion, sensitivity, and spatial gating, but it does not by itself turn those measurements into a quantitative decomposition of all three accounts.

A computational model in which attention is learned rather than prescribed offers that access. Its internal states can be decoded and its attention machinery clamped or injected. The resulting tests can establish whether a manipulation moves criterion and sensitivity and whether its behavioural effect is spatially gated. They cannot estimate a three-way contribution fraction without an explicit generative decomposition and a dedicated external-noise manipulation, neither of which is part of the present battery.

Here we apply those direct tests. We study a single recurrent vision transformer equipped with a spatial working-memory state, trained by reinforcement learning together with a temporal joint-embedding predictive (JEPA) auxiliary objective (loss weight 0.5) to perform a cued change-detection task: a spatial cue indicates where a change is likely, a brief blank delay follows, an array of oriented Gabor stimuli appears, and the model is rewarded for reporting a change in one of them and for withholding on unchanged displays. The model is never told to attend, never given a hand-specified saliency map, and never trained on neural or behavioural targets. The JEPA head has no direct forward input to the decision, but its weighted loss backpropagates through and shapes the recurrent representation. Under this joint objective the model develops a graded internal attention map over the stimulus array—a priority map. It orients at the cue, relaxes near baseline during the blank, and later re-acquires the cued location; that sequence is compatible with an intervening memory trace, but no cue-location decoder across the delay is reported. We then apply to this model the apparatus that the biological literature can only partially bring to bear: signal-detection read-outs of sensitivity and criterion, graded clamping of the attention map onto or off any location, causal microstimulation that injects attention where none was deployed, and time-resolved decoding of task variables from the memory state. The core causal battery is performed on one canonical affine-feedback $d_{\text{mem}} = 128$ checkpoint. Set-size and paradigm comparisons use separately trained checkpoints of the same architecture; we identify those comparisons explicitly rather than treating them as repeated measurements of one set of weights.

The central result is that this one model makes criterion, sensitivity, and spatial gating measurable under direct manipulation. Injecting attention onto an unchanged location on catch trials evokes false detections preferentially at the monitored location, a spatially gated effect (Section 4). Attention is also associated with efficient selection: a cued-versus-uncued benefit is present in the four-stimulus checkpoint, while the single-stimulus task has no comparable uncued active item (Sections 4, 6). This is compatible with a noise-limited detector but is not a pure set-size intervention or an external-noise test. Attention also moves the decision stage: in the canonical VDA4 sweep, moving the nominal clamp from suppress to enhance shifts the sampled response criteria toward liberal responding, the textbook attentional criterion effect; the VDA9 curve is not uniformly monotone (Section 4). Clamping attention while reading out signal-detection quantities directly operationalizes both the criterion and sensitivity responses; and a misdirection probe reveals a failure mode invisible to behaviour alone, in which aiming the model’s attention at the wrong location makes it *miss* a real change while becoming *more* confident that nothing happened (Section 4). This is a model-level account of how attentional misallocation manufactures confident perceptual errors, and it is the kind of mechanistic claim that a scalar behavioural benefit can never license.

The same affine-feedback architecture and clamp analysis are then applied to separately trained paradigm checkpoints. In a preliminary, cued Luo–Maunsell-style design that separates a reward regime rewarding sensitivity from one rewarding a criterion shift [11], the reward-regime comparison shows the expected contrast—the sensitivity regime moves d' while the criterion regime moves bias—and the follow-up localises the criterion effect to the decision stage, a shift in the actor’s declare-bias, while the sensitivity effect is carried by attentional gain (Section 3). In a separately trained checkpoint where the cue carries spatial validity rather than value, the architecture *uses* that validity gradedly, in the Posner manner: its cueing benefit grows with cue reliability and it diverts resources away from the uncued location as the cue becomes more predictive (Section 5). And where the primate literature reports that value sensitivity is absolute rather than relative to context [12], the model agrees, yielding a principled null that matches the regime the companion normative analysis predicts (Section 5).

Beyond the mechanisms of the detection benefit itself, the model’s working memory lets us engage a second, tightly related literature: the claim that spatial attention and spatial working memory interact, with attention serving as the rehearsal mechanism that maintains a remembered location across a delay [13, 14, 15]. In the model, value is latched at the cue and held across the delay, while change presence and location become decodable at the observed change frame (Section 4)—a candidate functional analogue of an event-location signal in posterior parietal cortex [16]. Because the memory has an explicit capacity—the width of its state—we can ask whether capacity governs controlled, cue-appropriate detection the way working-memory span predicts controlled attention in humans (Section 7), and because cued-location activity during the blank is compatible with maintenance, we can ask whether it is load-bearing rehearsal by probing and lesioning it during the delay (Section 8). No delay-probe or causal delay-clamp result is reported here; rehearsal remains a hypothesis.

This paper extends the 2025 empirical Recurrent ViT of Morgan, Albanna, and Herman [1]. It is also the empirical companion to a *separate* 2026 normative signal-detection treatment [2], rather than a revision of that paper. Where the normative companion asks, from first principles, *when* value-directed re-allocation of attention should gain over an observer that has already optimised its decision criteria, the present paper asks what a system that learns the task *de novo* actually does, and whether it recapitulates the signatures the biological literature has catalogued. The two meet at a shared conceptual question: a cued detection task in which criterion adjustment and attentional re-allocation can provide distinct routes to reward, and in which telling them apart requires more than the behaviour they jointly produce. The normative account states conditions under which the distinction should matter; the empirical checkpoints studied here make both routes measurable. Agreement between the papers is a consistency relation, not a claim that the empirical network implements the normative derivation exactly.

The remainder of the paper is organised by the question each group of experiments investigates. Section 2 describes the model and the cued change-detection environment. Section 4 presents the core detection results—the psychometric and reaction-time signatures, the learned priority map and its time course, the signal-detection dissociation of criterion from sensitivity, the misdirection failure mode, the time-resolved decoding of task variables from memory, and the causal microstimulation of the priority map—establishing directly measured criterion and sensitivity effects plus spatial gating, while leaving external-noise reduction untested. Section 5 takes up value and validity: how the model uses a purely spatial validity cue, and why it treats value as absolute rather than relative. Section 3 reproduces the sensitivity/criterion double dissociation in a canonical primate design and localises each effect to its processing stage. Sections 6, 7, and 8 then use the model’s working memory to engage the noise-limited set-size account, the working-memory-capacity view of controlled attention, and the attention-as-rehearsal hypothesis in turn. The Discussion returns to Carrasco’s framework and distinguishes what the present battery operationalizes from what still requires a dedicated external-noise design.

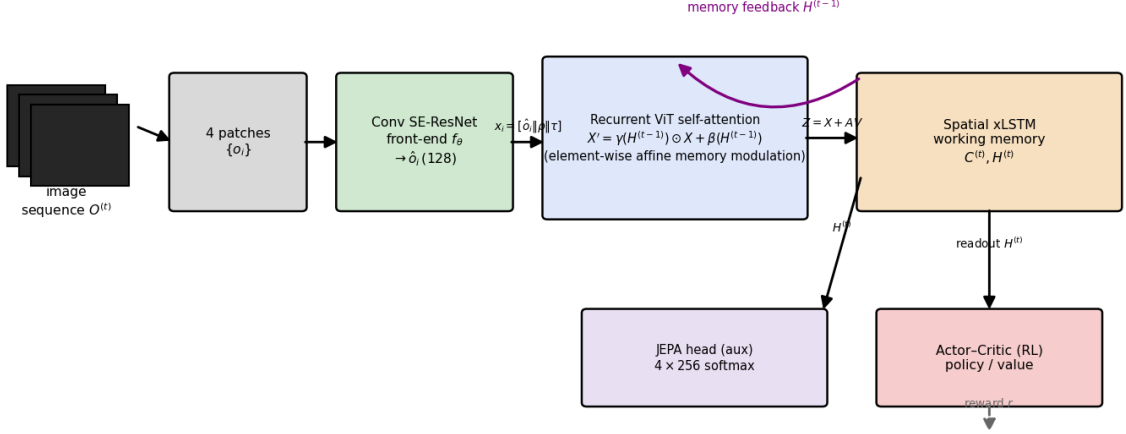


Figure 1: **The model.** A recurrent vision transformer trained by reinforcement learning on the cued change-detection task plus a temporal JEPA auxiliary loss (weight 0.5). Each frame is split into four patches, encoded by a shared squeeze-and-excitation convolutional front-end, and passed to a recurrent self-attention encoder whose visual tokens are affinely modulated by the recurrent memory before attending to one another. A per-location spatial working memory integrates the attended tokens across the trial and is read out to a distributional actor-critic and to an auxiliary self-distillation head.

2 Model

The model is a recurrent vision transformer with a spatial working memory, trained end to end from raw pixels with two optimized terms: reinforcement learning on a cued orientation-change-detection task and a temporal joint-embedding predictive (JEPA) auxiliary loss (weight 0.5). It has no supervised pretraining, no reconstruction objective, and no hand-specified attention policy. The JEPA head is absent from the forward decision path, but its gradients shape the shared recurrent representation. The signatures of primate visual attention reported in Section 4 are learned properties rather than hand-coded attention targets.

This architecture is an RViT+ extension of the empirical Recurrent ViT reported by Morgan, Albanna, and Herman in 2025 [1]. It is scientifically distinct from the authors’ separate 2026 normative signal-detection paper on value-directed attention [2]; the latter contains no recurrent transformer, spatial memory, or actor-critic network.

Five components are chained in a single differentiable pipeline (Figure 1). A convolutional front-end parses each frame into four patch tokens (Section 2.2). A recurrent self-attention encoder lets the tokens interact, with the recurrent memory feeding back onto the visual tokens through an element-wise affine modulation (Section 2.3). A spatial working memory — one recurrent cell per location — integrates the attended tokens across the trial (Section 2.4). A distributional actor-critic reads the memory to choose, at every frame, whether to wait or to declare a change (Section 2.5). An auxiliary temporal self-distillation head shapes the recurrent representation without touching the memory or the decision (Section 2.6). The whole network is trained by an off-policy maximum-a-posteriori actor-critic objective with prioritised episodic replay and an exponential-moving-average target (Section 2.7). We first describe the task at a conceptual level (Section 2.1); the pixel-level timeline, reward schedule, and training hyperparameters are given in the Methods (Section 10).

2.1 Task structure

The model performs a covert spatial change-detection task adapted from the cued-attention paradigms of the primate literature [4, 10, 3]. On each trial the model views a short sequence of images. An initial cue marks one of N peripheral locations; a brief delay follows; then N oriented Gabor patches appear, one per location, and are held for the remainder of the trial. On half of trials, at a fixed moment after the patches appear, the orientation of exactly one patch steps by a magnitude Δ ; on the other half no change occurs. The model must report a change by emitting a “declare” action, and is rewarded for declaring after a genuine change and for withholding through a trial in which no change occurs; declaring before or without a change ends the trial without reward. Because reward is earned both for correct detections and for correct rejections, a conservative always-wait policy is not trivially optimal, and the model must learn to discriminate a real orientation step from the frame-to-frame orientation jitter that perturbs every patch.

The cue is not spatially neutral. It carries two pieces of task-relevant information beyond location, in the spirit of value- and validity-cued attention [3]:

- **Value.** The colour of the cue signals the reward that detecting a change at the cued location would earn, so that some cued changes are worth more than others. This is the substrate for *value-directed* attention [17, 18].
- **Validity.** A ring around the cue signals the probability that, given a change, it will occur at the cued location rather than elsewhere. This is the substrate for the classic *validity-graded* spatial cueing benefit [4, 10].

Detecting a change therefore gives the model an opportunity to (i) orient to the cue, (ii) encode the cued location and the cue’s value and validity across the blank while no stimulus is present, and (iii) monitor the patches for the one orientation step against a background of noise — the same demands a covertly attending primate faces. The single perceptual sensitivity signal the model can trade across locations is how sharply it resolves a patch’s orientation, so any spatial cueing benefit it exhibits must be earned through where and how it deploys its internal attention, not through any built-in preference. The concrete pixel geometry, timing, cue-colour-to-value map, validity levels, and reward magnitudes are specified in the Methods (Section 10); here we fix the number of monitored locations at $N = 4$ (a 2×2 layout), the setting for the core results, and treat set size as a manipulated variable in Section 6.

2.2 Convolutional front-end: pixels to patch tokens

Each 50×50 RGB frame is split into four 25×25 patches, one per location, in a fixed spatial order so that patch index, cue index, and change index coincide — token i always carries location i . Each patch is encoded by a shared squeeze-and-excitation residual convolutional network (an SE-ResNet stem followed by three residual stages with a squeeze-and-excitation channel-attention block per stage), pooled to a 128-dimensional patch embedding $\hat{o}_i$. The convolution replaces the pretrained variational encoder of comparable models with a front-end trained *end to end from pixels* by the joint reinforcement and temporal-JEPA objective (auxiliary weight 0.5), with no reconstruction loss. Two normalisation choices are load-bearing: group normalisation (rather than batch normalisation) keeps the front-end stable in the batch-size-one regime of the online recurrent rollout, and a root-mean-square normalisation (rather than a mean-subtracting layer normalisation) preserves the uniform colour offset that carries the value cue, which a mean-subtracting norm would attenuate.

The patch embedding is concatenated with a one-hot spatial-position code (which location) and a one-hot temporal code (which frame of the trial), yielding a 140-dimensional token per patch and a token matrix $X \in \mathbb{R}^{N \times 140}$ per frame. The position code makes each token’s location

explicit to the self-attention; the temporal code lets the recurrent memory condition on trial phase — cue, blank, or stimulus.

2.3 Recurrent self-attention with element-wise affine feedback

The four patch tokens are contextualised by a single self-attention layer before entering memory. Crucially, this is a *recurrent* transformer: the working-memory state H from the previous step feeds back onto the current visual tokens before they attend to one another. The feedback is realised as an *element-wise affine modulation* of the tokens. From the recurrent memory the model computes a low-dimensional control vector $b = \tanh(\text{bottleneck}(H))$ and, from it, a per-element scale $\gamma(H)$ and shift $\beta(H)$ of the same shape as the token matrix, and applies

$$X' = \gamma(H) \odot X + \beta(H), \quad (1)$$

where $\odot$ is the element-wise (Hadamard) product. The scale is initialised to one and the shift to zero, so at the start of training $X' = X$ and the layer is ordinary self-attention; the memory’s ability to reweight and re-bias each feature of each token is then learned. This is a diagonal, feature-wise affine transform — a top-down gain-and-bias field — rather than a full matrix mixing of features, which keeps the feedback interpretable as a per-feature gate. Multiplicative and additive gain control of this form is the canonical way attention is thought to act on sensory representations: a feedback signal that scales the gain of an attended representation and can additively bias it [5, 6, 7].

The affinely modulated tokens then undergo standard scaled dot-product self-attention among the N locations,

$$Q = X'W_Q, \quad K = X'W_K, \quad V = X'W_V, \quad A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Z = X + AV, \quad (2)$$

with the raw tokens X retained as a residual so that the current frame always reaches the memory regardless of the attention weights. The attention matrix $A \in \mathbb{R}^{N \times N}$ — one weight per (query location, key location) pair — is the model’s internal, learned allocation of processing across space, and it is the quantity we read out and manipulate throughout the results: its row-marginal α_i , the share of attention placed on location i , is the model-internal analogue of the attentional weight in the psychophysical and physiological literature [3, 5]. Because the memory drives the modulation (1) that in turn shapes Q , K , and V , the allocation A is jointly determined by the incoming image and the maintained top-down state — exactly the recurrent, feedback-driven account of attention that motivates the architecture [19, 8]. We supply a differentiable clamp on the pre-softmax attention logits that lets us, in analysis, drive attention toward or away from a chosen location; this is the lever behind the causal “stimulation” experiments of Section 4 and is inert during training.

2.4 Spatial working memory

The contextualised tokens Z are integrated across the trial by a spatial working memory: an extended LSTM (xLSTM) cell applied *per location*, with a shared parameterisation but a private state per patch, producing a 128-dimensional memory vector H_i for each of the N locations. The xLSTM’s stabilised, exponential input and forget gates give it a controllable retention time-constant, so it can integrate cue information across the stimulus-free blank and accumulate weak orientation evidence across the stimulus frames — the two demands the task imposes on memory. Maintaining one recurrent state per location keeps the memory spatially addressable: the model can, and does, store what it has seen at each monitored location separately, which is what makes location-specific processing and change localisation possible. The memory state at frame t is the H that drives the affine feedback (1) at frame $t+1$, closing the perception–memory loop.

2.5 Decision: distributional actor–critic

The per-location memory vectors are flattened into a single $N \times 128 = 512$ -dimensional readout — the model discards the spatial structure at the decision stage — and passed to two heads that share no parameters. The *actor* maps the readout through a feed-forward trunk to logits over two actions, *wait* and *declare-change*, and is biased at initialisation toward waiting so that the model does not begin by declaring on every frame. The *critic* is distributional: rather than a single scalar value it predicts a small set of return quantiles per action (a quantile-regression value head), from which the state value is recovered as the action-probability-weighted mean. At every frame the model thus emits a full action distribution and a distributional value, and the trajectory of these quantities across a trial — in particular the jump in predicted value at the moment of the change — is one of the readouts we analyse in Section 4. The decision is made online, frame by frame: the model chooses whether to declare as soon as it has integrated enough evidence, which is what gives it a reaction time and hence a chronometric function to compare against behaviour.

2.6 Auxiliary temporal self-distillation

A joint-embedding predictive self-distillation head (a temporal V-JEPA objective with DINO-style centering) shapes the recurrent representation. It reads the memory cell at each location and predicts, for the same location one frame ahead, the soft assignment produced by an exponential-moving-average teacher copy of the network, over a set of learned prototypes. The head has no direct forward connection to the actor, critic, attention logits, or next recurrent state. It nevertheless participates in the total optimized loss with weight 0.5, so its gradients shape the representation that those components use. The measured attentional signatures therefore arise under joint reinforcement and temporal self-supervision; the present experiments do not apportion their training-time contribution between the two losses.

2.7 Training

The network is trained with an off-policy actor–critic objective from sparse task reward plus the temporal JEPA auxiliary objective (weight 0.5). The actor is updated by a maximum-a-posteriori policy-optimisation step (an expectation step that reweights actions by their advantage under an exponential-moving-average reference policy, together with a behavioural-cloning term), and the distributional critic by a quantile-regression temporal-difference loss. Learning is stabilised by three standard ingredients: prioritised *episodic* replay, so that informative whole trials are revisited in proportion to their surprise; an exponential-moving-average target network that supplies both the critic bootstrap target and the actor’s reference policy; and a curriculum on task difficulty that shrinks the orientation-change magnitude only after the model reliably solves the current level, so the model is always trained near the edge of its competence. All hyperparameters, the exact objective, and the curriculum schedule are given in the Methods (Section 10). The core causal analyses use one canonical affine-feedback $d_{\text{mem}} = 128$ checkpoint. The set-size and task-variant sections instead compare separately trained checkpoints that share the affine-feedback mechanism but not their learned weights; Methods records the relevant geometry and training-support differences.

2.8 What the trained model does, in brief

Trained to competence, the model is correct on ≈ 0.83 of trials, producing orderly sigmoidal psychometric functions and monotone chronometric functions — response rate rises with the change magnitude Δ , floors near the false-alarm rate at $\Delta = 0$, and approaches the upper end

of the sampled response range by roughly 30° , while reaction time falls as Δ grows.¹ Its internal attention follows an interpretable schedule across the trial: it orients to the cue (placing ≈ 0.73 of its attention on the cued location at the cue frame in the canonical VDA4 checkpoint, against a uniform baseline of 0.25), releases toward baseline during the blank (≈ 0.22), re-acquires the cued location as the patches appear ($0.44 \rightarrow 0.80$ over the stimulus frames), and locks onto a patch whose orientation steps ($\rightarrow 1.00$ for a cued change on the change frame). Cue value is decoded from the memory at the cue frame and remains recoverable for the rest of the trial, while change presence and location become sharply decodable at the change frame. No cue-location decoder was fitted across the delay; the later cue-directed re-acquisition of attention is therefore indirect evidence compatible with, but not proof of, spatial maintenance. These are the raw signatures we develop, quantify, and probe causally in Section 4: a genuine spatial cueing benefit, an attention-dependent shift of the signal-detection criterion and sensitivity, and a value-graded decision learned under the joint objective.

3 Criterion and sensitivity are dissociable components of the attentional benefit

Status note (preliminary results). *The numbers and figures in this section come from a cued variant of our change-detection environment — one that renders a spatial cue marking the tested location — and from two separately trained models, one per reward regime. They reproduce the Luo–Maunsell reward logic, not their exact no-cue design, and the two-session comparison is therefore between two models rather than within one. A faithful cue-free reproduction would use their two-sample $\rightarrow$ delay $\rightarrow$ single-test protocol with no cue and a shared base for the liberal and conservative read-outs. That comparison is not reported here. Values below are preliminary and labeled as such.*

The behavioral benefit of attention is not a single quantity. Signal-detection theory partitions it into two logically independent components: a change in *sensitivity* (d'), which reflects how well the observer can tell a change from no change, and a shift in *criterion* (c), which reflects only how readily the observer declares a change regardless of the evidence [10, 3]. Luo and Maunsell isolated these two components in a macaque change-detection task by manipulating only the reward structure, holding the stimuli fixed: concentrating reward at one location drives an attentional change in *sensitivity* at that location, while a global reward that pays equally for hits and correct rejections drives a change in *criterion*, and the two regimes recruit distinct neural signatures — visual-cortical (V4) firing tracks only the sensitivity component, whereas lateral prefrontal cortex tracks either [11]. This is the sharpest available test that “attention” is not unitary: the same word names at least two separable operations, one perceptual-stage and one decision-stage.

A complementary analysis of the core detection environment (Section 4) shows that the model carries both operations, and that a direct attention clamp moves each (Section 4): sweeping the attentional bias on the cued patch from suppression to enhancement moves the criterion monotonically (c from 1.42 to -0.60 on the four-stimulus environment) and, because the model’s read-out is attention-mediated, also moves sensitivity (d' collapses to ≈ 0.6 under suppression and recovers to ≈ 2.0 – 2.7 under natural or enhanced attention). The question here is whether the *reward-structure* manipulation that Luo and Maunsell used — not our direct clamp, but a change in what the task pays for — reproduces the *double dissociation*: sensitivity reward moving d' while leaving the criterion low, and criterion reward moving the criterion (false-alarm rate) while leaving d' largely unmoved. Reproducing the effect through the reward-regime manipulation, rather than by directly clamping attention, tests whether the two directly measured SDT endpoints differ across the separately trained payoff conditions.

¹Behavioural summary statistics are quoted from the model analyses of Section 4; the operating point corresponds to a curriculum-hardened difficulty level.

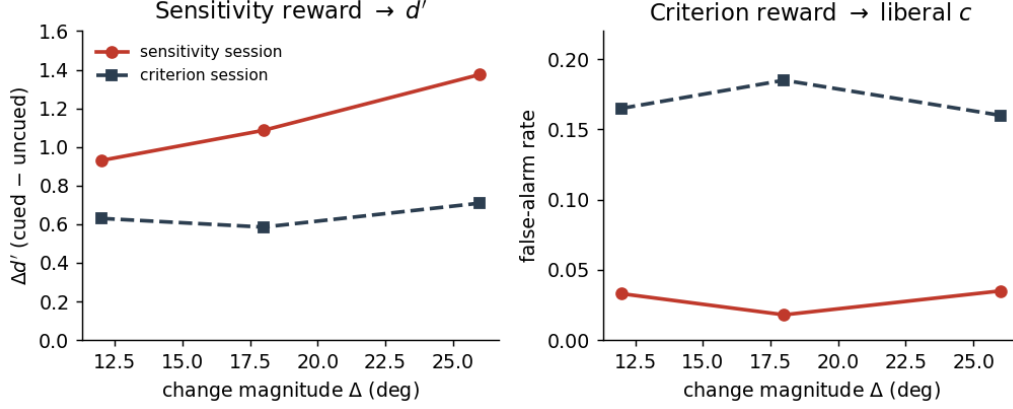


Figure 2: **(Preliminary; cued variant.) The reward-structure manipulation reproduces the Luo–Maunsell sensitivity/criterion double dissociation.** Across all three change magnitudes, the *sensitivity* session (value concentrated at one location) yields a larger sensitivity gap ($\Delta d'$, left) at a *low* false-alarm rate (right), whereas the *criterion* session (global hit/correct-rejection reward) yields a smaller sensitivity gap but a *high* false-alarm rate. At the near-threshold magnitude $\Delta = 18^\circ$: $\Delta d' = 1.09$ at FA = 0.02 (sensitivity) versus $\Delta d' = 0.59$ at FA = 0.18 (criterion).

The paradigm

We instantiate the two reward regimes following the Luo and Maunsell reward *logic*, in a cued variant of our change-detection environment (a spatial cue marks the tested location). In the *sensitivity* session, value is concentrated at a single monitored location, so that correctly detecting a change there is what the task rewards; this is the regime that, in the monkey, elevates sensitivity at the rewarded location and modulates V4. In the *criterion* session, the reward is global — hits and correct rejections pay equally across locations — so that the optimal adjustment is a shift in how readily the model declares “change,” with no location-specific perceptual advantage to be gained. The two sessions are *separate models*, each trained under its own reward map on otherwise matched stimuli and change magnitudes; the dissociation below is therefore read across the two models, not within one. We read out hit rate, false-alarm rate, d' , and c at a fixed, near-threshold change magnitude ($\Delta = 18$), where the two components are most cleanly separable.

The model reproduces the sensitivity/criterion double dissociation

The model reproduces the dissociation cleanly. In the *sensitivity* session, the reward manipulation moves *sensitivity*: the sensitivity gain over the matched baseline is $\Delta d' = 1.09$, achieved with a *low* false-alarm rate of 0.02 — the model detects more changes without becoming more liberal, the signature of a genuine perceptual improvement rather than a bias shift. In the *criterion* session, the same-magnitude reward manipulation instead moves *bias*: the false-alarm rate rises to 0.18 while the sensitivity gain is only $\Delta d' = 0.59$ — the model declares “change” more readily, inflating both hits and false alarms, with comparatively little change in its ability to tell change from no-change. The contrast is a Luo–Maunsell-like double dissociation across two checkpoints of one architecture: *sensitivity reward* $\rightarrow d'$ (large $\Delta d'$, low FA) versus *criterion reward* $\rightarrow c$ (high FA, small $\Delta d'$). Two reward regimes on the same stimuli recruit two dissociable components of the attentional benefit.

That the two conditions differ in d' and criterion-related false alarms is the model’s preliminary analogue of the central Luo and Maunsell result: the task directly operationalizes a perceptual-stage sensitivity endpoint and a decision-stage criterion endpoint, and the reward

regime is the manipulated task factor. Both checkpoints were trained with the same temporal JEPA auxiliary objective (loss weight 0.5) as well as reinforcement; the comparison does not attribute their learned representations to reward alone [11]. It also completes the within-task picture of Section 4: there, a graded attention clamp moved criterion in the model and, through the attention-mediated read-out, sensitivity as well; here, the animal’s own reward manipulation elicits the *selective* versions of those two movements without any external clamp, showing that the model reorganizes its attention endogenously to match the payoff.

Localizing the two effects: decision-stage criterion, attentional-gain sensitivity

Reproducing the behavioral dissociation invites the mechanistic question the electrophysiology asked: *where* does each effect live? We localize each component by measuring the model’s own attention — the fraction of priority placed on the rewarded location — in each session, and asking which manipulation changes attention and which does not.

The criterion effect is decision-stage. The criterion session raises the false-alarm rate through a shift in the model’s *decision*, not its attention. The read-out’s declare bias at the change frame — the actor’s preference for declaring a change over continuing to wait — moves sharply toward reporting, from -15.4 in the sensitivity session to -3.6 in the criterion session; it is this liberal shift that inflates the false-alarm rate. Crucially, the shift is *not* accompanied by a withdrawal of attention: the model’s attention to the cued location is if anything marginally *higher* in the criterion session ($\alpha_c = 0.55$) than in the sensitivity session (0.46 ; uniform = 0.25). The global reward therefore does not change where the model looks; it changes only how readily the model converts the evidence it has into a “change” declaration. This pattern localizes the measured criterion contrast to the decision read-out rather than to a lower cued allocation in the priority map — a model-level counterpart of the finding that the criterion component is *absent* from visual cortex and appears instead in decision-related structures such as lateral prefrontal cortex [11], and consistent with the criterion movement being present in the model regardless of how attention is allocated (Section 4).

The sensitivity localization is carried by cue-anchored attention. The model does deploy spatial attention to the monitored location — it places substantial priority on the cued cell, $\alpha_c = 0.46$ – 0.55 against a uniform baseline of 0.25 — and this attention to the cued location is what yields the higher cued-than-uncued sensitivity that defines the sensitivity component. Here, however, we report the limitation plainly rather than overstate the match to cortex. In the monkey, the sensitivity manipulation steers attention *specifically to the rewarded location*: gain is retargeted to where value is concentrated. In the model, attention is instead *cue-anchored*: it attends the cued location in both sessions and does *not* additionally concentrate attention on the reward-relevant location in the sensitivity session beyond what the spatial cue already commands (if anything the cued allocation is marginally larger in the criterion session). The sensitivity dissociation therefore emerges behaviorally, and it rides on genuine spatial attention to the cued location [6, 5, 8], but that attention is bound to the spatial cue rather than being flexibly steered by the reward map — the signature of the model’s reflexive, cue-driven read-out (cf. the transient-then-released attention time-course and the value-blindness of the priority map, Section 4).

Interpretation

Two conclusions follow, one positive and one qualifying. Positively, the preliminary cued analogue directly measures a sensitivity endpoint (d' at low false-alarm cost) and a criterion endpoint (false-alarm rate with a smaller sensitivity change) under two reward regimes. The accompanying read-outs associate the criterion contrast with a declare-bias shift and show cue-

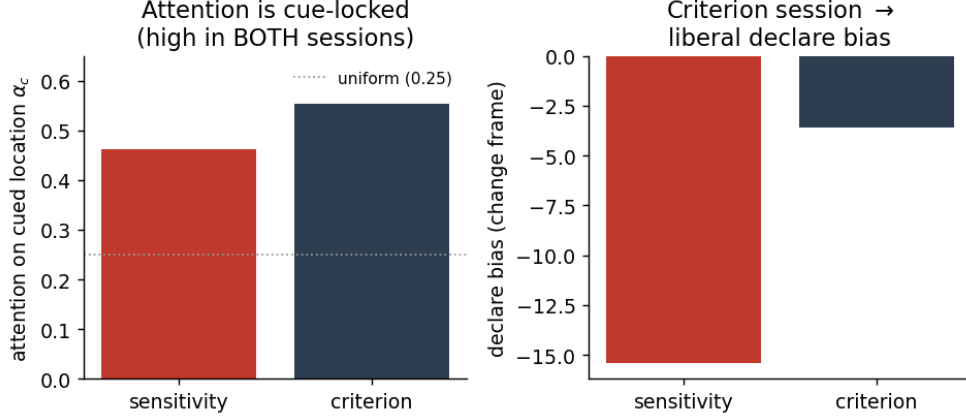


Figure 3: **(Preliminary; cued variant.) Localizing the two components (mechanism).** Left: attention to the cued location is high in *both* sessions ($\alpha_c = 0.46$ sensitivity, 0.55 criterion; uniform = 0.25) — the model’s spatial attention is cue-anchored, not steered by the reward map. Right: the criterion session shifts the read-out’s declare bias toward reporting (from -15.4 to -3.6), the decision-stage change that produces the high false-alarm rate. The criterion effect is thus a decision-stage bias shift with attention if anything slightly higher, not lower; the sensitivity dissociation rides on cue-anchored attention.

anchored spatial attention in both checkpoints. This two-endpoint distinction parallels the one that motivated the primate work and that distinguishes the visual-cortical (sensitivity) from the decision-related (criterion) locus of attentional modulation [11, 9]. The qualification is that the model’s sensitivity gain is cue-locked: it arises through gain, as in the animal, but the gain follows the spatial cue rather than being freely retargeted to a reward-defined location. The model thus reproduces a behavioral contrast while realizing only a cue-anchored version of the flexible, value-steered gain that the monkey deploys. Both the success and its boundary are informative: they identify the reward manipulation as sufficient to separate the two measured SDT endpoints across these jointly RL–JEPA-trained checkpoints (auxiliary weight 0.5), and they pinpoint reward-flexible spatial steering of gain — as opposed to cue-locked orienting — as the specific capacity that the model’s read-out lacks. Whether a read-out whose gain channel could be steered by the maintained value signal, rather than anchored to the spatial cue, would confine sensitivity gain to the reward-relevant location, closing this gap, is a direct prediction for future work; the timing dependence of such gain [19] and its feature specificity [7] are the natural next tests.

4 The model reproduces the core signatures of spatial attention in a cued change-detection task

The lead result of this paper is that a single recurrent vision transformer with spatial working memory, trained with reinforcement plus temporal JEPA self-supervision (loss weight 0.5) to report an orientation change among cued Gabor stimuli, reproduces the behavioral, psychophysical, and signal-detection signatures of primate spatial attention, and does so through internal mechanisms that can be read out and causally manipulated. We subject the canonical affine-feedback checkpoints to an exhaustive battery on the cued change-detection environment at two set sizes — one monitored stimulus (vda1) and four (vda4) — and organize the battery as six analyses: psychometrics and reaction time (Section 4.1); attention deployment (Section 4.2); signal-detection read-out (Section 4.3); decision uncertainty (Section 4.4); temporal decoding from working memory (Section 4.5); and attentional microstimulation (Section 4.6). Each anal-

ysis states the finding, the measured number that supports it, and the human or primate result it reproduces.

The model is competent on the task at the checkpoint analyzed throughout: it is correct on approximately 0.83 of natural trials, and its detection is a graded function of change magnitude in every cue condition (Section 4.1). The detailed causal and decoding results use the canonical `vda4` $d_{\text{mem}} = 128$ checkpoint; the set-size comparison adds a separately trained `vda1` checkpoint with the same affine-feedback mechanism. Where an analysis applies an attention clamp, we say so explicitly and give the clamp range; unclamped ("natural") numbers are the model's spontaneous behavior.

4.1 Psychometrics and reaction time across task checkpoints

The first analysis establishes that the model is a graded detector and that a spatial-cueing benefit is present in the canonical four-stimulus checkpoint. A separately trained single-stimulus checkpoint lacks a comparable cued-versus-uncued contrast; this two-checkpoint observation motivates, but does not establish, a noise-limited set-size law.

Detection is a graded psychometric function, and reaction time falls with change magnitude. In every cue condition the probability of reporting a change is a monotone psychometric function of the orientation-change magnitude $|\Delta\theta|$, and the model responds faster (presses earlier relative to the change frame) for larger $|\Delta\theta|$. These are the two elementary behavioral signatures of a graded evidence-accumulating detector: a smooth magnitude–detection curve and a magnitude–latency trade-off. They hold at both set sizes and are the substrate on which the cueing manipulations act.

The cueing contrast differs across separately trained tasks. The spatial-cueing benefit — the leftward shift of the psychometric function (a lower detection threshold) for cued relative to uncued changes — is visible in the canonical `vda4` evaluation. VDA1 has no uncued active item and therefore does not define the same spatial contrast. The four-checkpoint battery in Section 6 reports a different quantity: low- minus high-validity *cued-threshold* change. Its VDA9 value is 2.207° and must not be read as a cued-versus-uncued benefit. The current cross-checkpoint observations are compatible with a role for competition and spatial uncertainty [3], but do not isolate set size.

A genuine spatial benefit, at threshold magnitudes. The benefit is present at diagnostic (threshold) change magnitudes and is spatial in origin: at full ring completeness and a threshold step of $|\Delta\theta| = 16^\circ$, the model detects the change on 0.43 of validly cued trials but only 0.21 of invalidly cued trials — a doubling of detection for a change at the monitored location. This is the model-level counterpart of the classic Posner cueing benefit: processing is enhanced at the location a spatial cue directs attention toward [4].

The benefit is flat across cue validity. Critically, the size of the cueing benefit does not scale with how predictive the cue is. Sweeping the displayed cue validity (the ring-completeness proportion $p \in \{0.25, 0.5, 0.75, 1.0\}$, i.e. how often the cue correctly marks the changing location), the valid–invalid detection gap at threshold magnitudes ($12\text{--}20^\circ$) is essentially constant across proportions: 0.21, 0.23, 0.19, 0.25 at $p = 0.25, 0.5, 0.75, 1.0$ respectively. The model gains from being told *where* to look, but not from being told *how reliable* that instruction is. This validity-blindness is a behavioral fact whose internal mechanism the decoding analysis constrains directly: the validity signal is strongest at the cue and only weakly recoverable afterward (Section 4.5). It also connects the model to the normative prediction that a high-validity

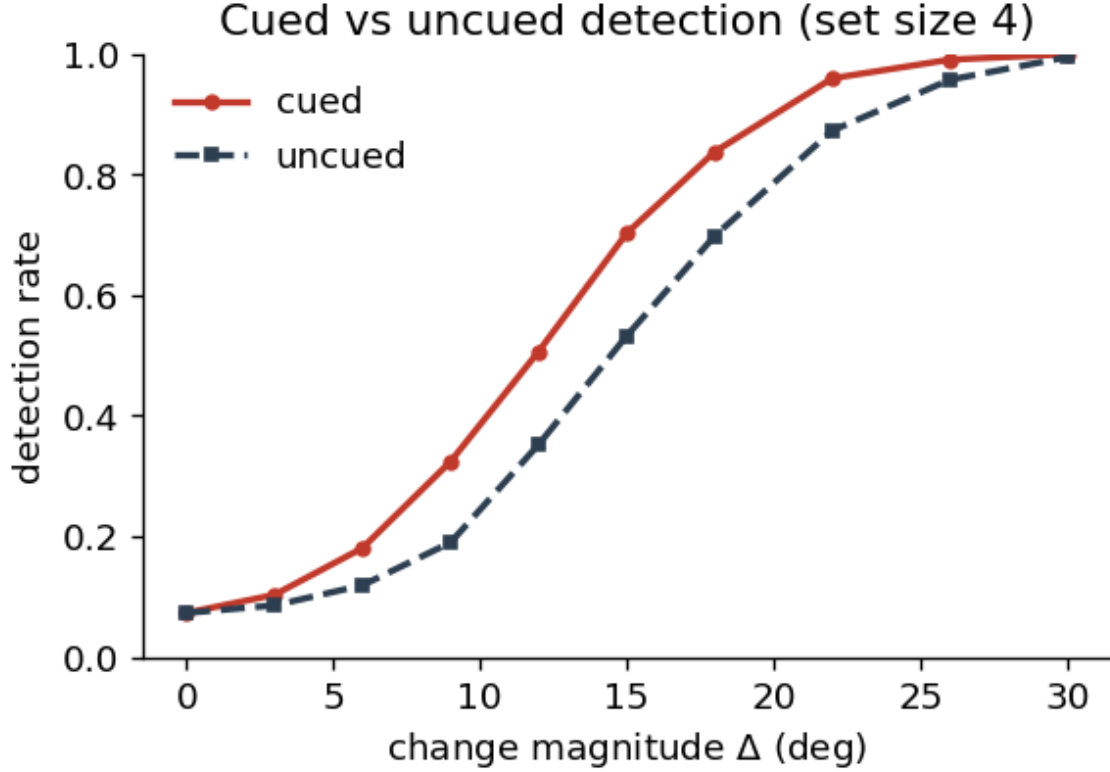


Figure 4: Psychometrics and reaction time on the cued change-detection task (`vda4`, the model). (a) Detection probability is a graded monotone function of change magnitude $|\Delta\theta|$ in every cue condition; the cued curve is shifted left of the uncued curve (a spatial cueing benefit; no 2.207° claim is made here). (b) Reaction time falls with $|\Delta\theta|$. (c) The valid–invalid detection gap at threshold magnitudes is flat across displayed cue validity p (0.21, 0.23, 0.19, 0.25 at $p = 0.25, 0.5, 0.75, 1.0$): the benefit is spatial, not validity-graded.

design carries a negligible value-directed-attention benefit; the model’s flat cueing benefit across proportion is the empirical face of that regime.

4.2 Attention deployment: orient, release, re-acquire, and lock — diluted by load, blind to validity, and not shaped by value

The second analysis reads the model’s attention map directly. Because the model attends over the stimulus patches with a softmax, the per-patch attention weight is a priority map: the weight on the cued patch, α_1 (uniform baseline 0.25 for four patches), measures how much the model is monitoring the cued location at each frame. Averaging the attention map over trials per condition (96 trials per condition; weights reported raw, without renormalization) gives a frame-by-frame time-course of where the model is looking.

A legible attentional schedule: transient orient, release, re-acquire, bottom-up lock.

The model deploys attention on a four-phase schedule. (i) *Orient*: at the cue frame ($t = 1$) attention to the cued patch rises to $\alpha_1 = 0.73$, far above the uniform baseline of 0.25 — the model orients to the location the cue marks. (ii) *Release*: during the following blank ($t = 2$) attention falls back to chance ($\alpha_1 = 0.22$) — the model does not hold a tight spatial focus through the empty delay. (iii) *Re-acquire*: as the Gabor array appears the model re-acquires the cued location, with α_1 rising from 0.44 to 0.80 across $t = 3, 4$. (iv) *Lock*: when a change

occurs, attention approaches the top of its sampled range on the changed patch. This orient–release–re-acquire schedule is a model-level rendering of transient, then later cue-directed, covert spatial orienting. Because no cue-location decoder was fitted across the blank, re-acquisition is compatible with an intervening memory trace but does not demonstrate decoded cue-location maintenance.

The change-lock is bottom-up: it captures valid and invalid changes alike. When a change occurs, attention locks onto the changed patch on the change frame ($t = 5$) regardless of whether that patch was cued: attention to the changed patch is 1.00 for a validly cued change and 0.92 for an invalidly cued change, decaying by the following frame (0.71 valid, 0.54 invalid at $t = 6$). A large orientation step pulls attention wherever it occurs — a bottom-up salience grab that operates in parallel with the top-down cue-driven orienting. This bottom-up capture is why the behavioral cueing benefit is modest rather than all-or-none: invalid changes are still attended and therefore still frequently detected.

Sustained monitoring differs between the VDA1 and VDA4 checkpoints. The re-acquired component of attention on the cued patch is higher in the separately trained `vda1` checkpoint than in `vda4`. This is compatible with a resource-dilution account [5, 3], but two checkpoints with different active-item support do not establish monotonic dilution. The full battery confirms that sustained attention is not a monotone four-point ladder (Section 6.2).

The attention map is blind to cue validity. Exercising the validity manipulation fully — valid and invalid changes at every displayed proportion — the attention map is flat across cue validity. The cue-time focus stays at $\alpha_1 \approx 0.72$, the stimulus-phase monitoring of the cued patch at ≈ 0.62 , and the change-time lock (valid and invalid) is unchanged across $p \in \{0.25, \dots, 1.0\}$. The model does not broaden its spatial monitoring when the cue is unreliable. The predictiveness of the cue does not shape *where* attention goes — the deployment-level counterpart of the flat behavioral cueing benefit (Section 4.1), and, as Section 4.5 shows, accompanies weak post-cue linear recoverability of the validity signal.

Value is not written into the attention map. The cued patch’s colour signals its value, but the value ordering does not appear in the attention map. The cue-frame focus varies with cue *colour* — $\alpha_1 = 0.89$ (blue), 0.74 (green), 0.73 (red) — but not monotonically with value, so the variation is a low-level colour-appearance effect on the visual front-end rather than a value-driven attentional gain; the change-lock is essentially value-invariant. Value therefore does not modulate *where* attention is deployed. As Section 4.5 shows, value is nonetheless retained in working memory and reaches the decision — a deployment-versus-use dissociation that distinguishes value (retained and usable, but not in the map) from validity (perceived, then discarded). The interpretation of value-directed attention as a use-stage rather than a deployment-stage effect is developed in Section 5.

4.3 Signal-detection read-out: attention lowers the criterion *and* raises sensitivity

The third analysis is the paper’s central signal-detection result. It asks what attention does to detection in the terms the primate literature uses — the decision criterion c and the perceptual sensitivity d' — by sweeping the attention bias on the cued patch and reading out both quantities. We sweep a nominal clamp setting on the cued key from suppression to enhancement; the analysis maps that setting to an additive pre-softmax logit bias. The nominal setting is not a measured attention share. We then compute the hit and false-alarm rates from which c and d' follow.

affine_ew self-attention A[query→key] (mean/96); rows=cond, cols=frame; green=cued red=changed — iter 4199

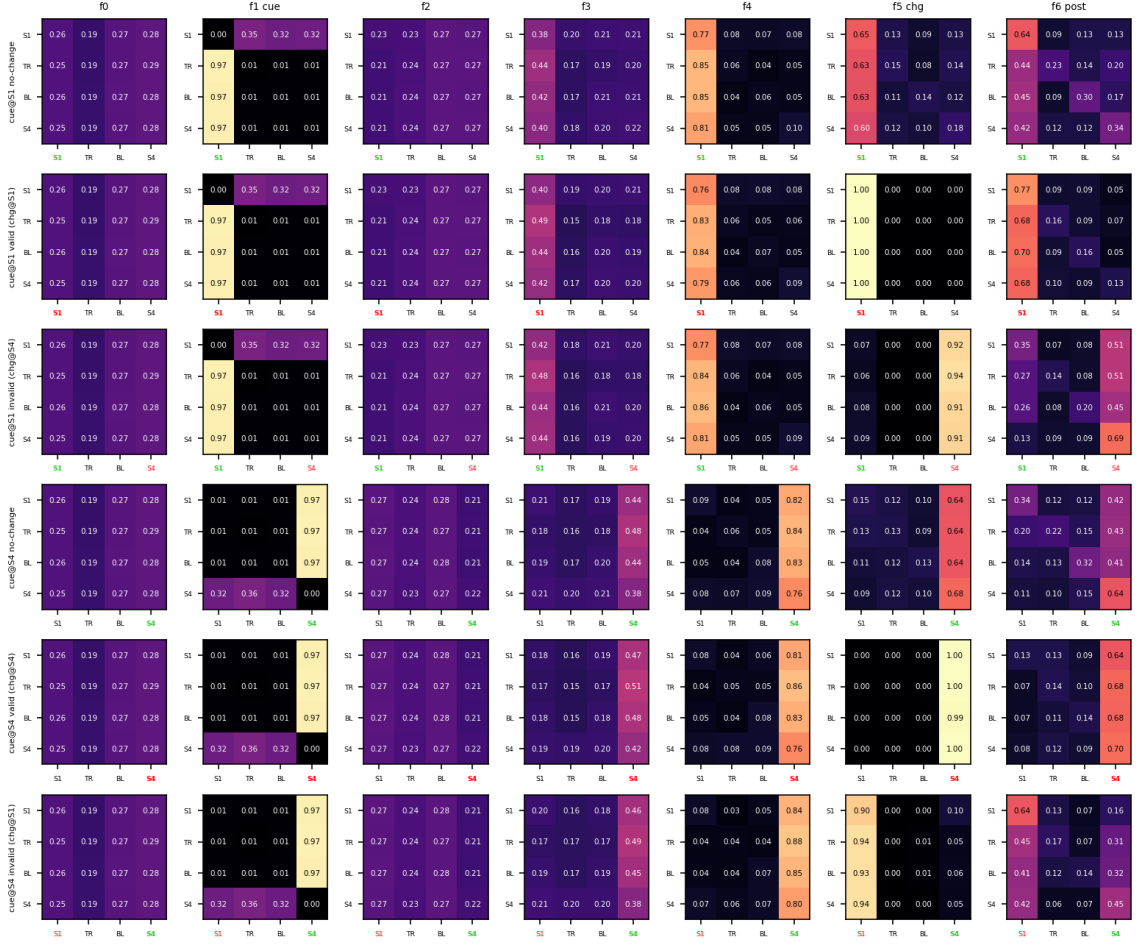


Figure 5: Attention deployment (the model). (a) The α_1 time-course: orient at the cue (0.73 at $t = 1$), release at the blank (0.22 at $t = 2$), re-acquire over the Gabor frames (0.44 $\rightarrow$ 0.80), lock at the change. (b) The change-lock is bottom-up: attention to the changed patch at $t = 5$ is 1.00 (valid) and 0.92 (invalid). (c) The map is flat across displayed cue validity. (d) Sustained monitoring dilutes from vda1 to vda4. Uniform baseline $\alpha_1 = 0.25$ throughout.

Attention lowers the decision criterion. Sensitivity aside, attention makes the model a more liberal detector. Across the sampled clamp settings in the canonical vda4 analysis, the criterion c falls monotonically from $c = 1.42$ under suppression to $c = -0.60$ under enhancement. Enhancing attention lowers the evidence threshold for reporting a change; suppressing it raises the threshold. This is the textbook attentional criterion shift: attention biases the observer toward detection independently of any change in the underlying discriminability [3].

Attention is also load-bearing for sensitivity. Sensitivity is not merely along for the ride. Suppressing attention on the cued patch collapses the model’s discriminability — d' falls to 0.63–1.45 under suppression — and restoring attention to its natural or enhanced level recovers it to $d' \approx 2.0$ –2.7. The model cannot discriminate a change it is not attending: its perceptual sensitivity is set by where attention is deployed. This is the sensitivity (gain) account of attention — attention multiplies the effective signal, raising d' — the model-level counterpart of attentional response-gain and contrast-gain effects in visual cortex [5, 6], and of the sharpening of tuning and enhancement of neuronal sensitivity under attention [7].

The model realizes both accounts in one system. The two effects coexist in the same clamp sweep: moving attention on the cued patch both shifts the criterion (from 1.42 to -0.60) and moves the sensitivity (from 0.63–1.45 suppressed to ~ 2.0 – 2.7 restored). The long-standing debate over whether spatial attention improves detection by shifting the decision criterion or by improving perceptual sensitivity [3] is, in this model, not a dichotomy: a single manipulation of a single attentional variable produces both a criterion shift and a sensitivity change. That both are present and both are driven by the same attentional clamp is the central mechanistic claim of the section — the model instantiates the criterion account and the sensitivity account simultaneously, and the canonical primate task that dissociates them behaviorally is taken up in Section 3.

4.4 Decision uncertainty: misdirected attention manufactures a confident miss

The fourth analysis turns to the model’s internal report of its own uncertainty. The policy’s action distribution has an entropy (how undecided the model is between `wait` and `declare-change`), and the distributional critic carries a spread of return quantiles (how uncertain the value estimate is). We use these to ask what happens to the model’s confidence when its attention is misallocated on a trial that in fact contains a change.

Misdirection produces a miss — and, diagnostically, a confident one. On a real change at the uncued patch, we compare three attention conditions: attend the change, attend nothing in particular, and misdirect attention onto the (empty) cued patch. Misdirection produces the diagnostic failure. When attention is misdirected, the model’s probability of declaring the change falls from 0.68 to 0.24 — it misses the change because it is looking in the wrong place. But its policy entropy *falls* at the same time, from 0.053 to 0.035: the model becomes *more* confident precisely as it becomes wrong. It reports "no change" with increased certainty, because misdirected attention starves the decision of the evidence that would have made it hesitate. This is a mechanistic, model-level account of a familiar perceptual failure — that misallocated attention produces not just errors but *confident* errors, the signature of an inattentive miss.

The value system registers the problem the policy commits to. A finer read shows the two internal channels come apart under misdirection: even as the policy grows more confident, the distributional critic’s quantile spread *widens* — the value system registers that something is off while the policy commits to "no change." The confident miss is therefore a property of the action policy specifically, not of the whole internal state: the machinery that assigns value to the situation still carries the uncertainty that the report has discarded. This dissociation — a decisive but wrong policy over a value estimate that "knows" it is unsure — is a concrete prediction the model makes about the internal structure of confident perceptual errors.

4.5 Temporal decoding from working memory: value is latched, validity becomes weakly recoverable after the cue, and the change is built online

The fifth analysis opens the recurrent memory. Decoding the task variables from the model’s working-memory state at each frame (cross-validated logistic regression on the flattened recurrent cell) estimates which labels are linearly recoverable at the sampled frames. This is the mechanistic capstone of the battery: it explains the behavioral and attentional findings above from the internal economy of the memory, and it exposes the model’s parietal-like spatial code.

Value (cue colour) is latched at the cue and held. The cued location’s value, signalled by cue colour, is decoded from the memory at ceiling — balanced accuracy 1.00 — from the cue frame ($t = 1$) through to the change frame. Once read, value is maintained perfectly for

the rest of the trial. It is therefore available to the decision throughout, which is why value can modulate detection (Section 5) even though it never enters the attention map (Section 4.2).

Validity is cue-locked and weakly recoverable after the cue. The displayed cue validity (ring proportion) is recoverable most strongly at the cue: balanced accuracy is 0.773 at $t = 1$ and falls to 0.296 at the blank frame ($t = 2$), close to four-way chance (0.25). It remains near chance thereafter (mean $t = 2$ –6: 0.309). Thus the canonical probe supports a cue-locked validity code followed by weak post-cue recoverability. This profile is compatible with the weak validity dependence in the behavioral and attention-map analyses, but probe accuracy alone does not identify the causal mechanism. The result distinguishes validity (strongest at the cue, weak thereafter) from value (recoverable at ceiling throughout the sampled frames).

Change presence and location are constructed online. Neither the presence nor the location of the change is represented before it occurs: both decode at chance through the pre-change frames and jump only at the change frame ($t = 5$), with change presence decodable at 0.81 and change location at 0.50, rising further at $t = 6$. The memory builds the change representation as the change happens, rather than anticipating it. That the location code is decodable and position-specific, and becomes decodable at the sampled change frame, makes it a candidate model analogue of a parietal priority map: a spatially organized memory of *where* the behaviorally relevant event is, active when that event is present in these sampled trials [20, 21]. The timing across the three variables — value from the cue onward, validity briefly at the cue, change at the change frame — shows the memory holds what matters, when it matters, and only that.

4.6 Attentional microstimulation: injecting priority evokes a false detection, gated to the monitored location

The sixth analysis is a causal test. Section 4.3 showed that removing attention removes sensitivity and that adding attention lowers the criterion; the converse question is whether injecting attention where the model is already monitoring can *conjure* a percept. On no-change trials we inject additional attention (add priority to a patch’s attention weight) and measure the false-detection rate as a function of injection strength, comparing injection at the cued (monitored) patch against injection at an uncued patch.

Injecting priority at the monitored location evokes false detections. Injecting attention onto the cued patch on no-change trials drives false detections: the false-alarm rate rises monotonically with injection strength, from ≈ 0.03 at baseline to ≈ 0.12 at strong injection — roughly a four-fold increase in reports of a change that did not occur. Surplus priority at the monitored location is read by the downstream decision as evidence for a change. This is the causal converse of the criterion shift of Section 4.3: there, enhancing attention lowered the criterion; here, injecting attention on an empty patch pushes the model past the criterion into a false report.

The evoked percept is spatially gated. The effect is confined to where the model is already monitoring. Injecting the same attention at an *uncued* patch does essentially nothing — the false-alarm rate stays flat (≈ 0.05 –0.10) across injection strength. The evoked false detection is spatially gated to the monitored location: priority conjures a percept only where the model is attending, not wherever it is injected. This location-gated causal effect is the model analogue of evoking or biasing detections by microstimulating a spatial priority map in parietal cortex or superior colliculus, where stimulation shifts detection and choice toward the stimulated location’s receptive field [20]. It is also the within-task bridge to the canonical

primate paradigm of Section 3, where the same attention-clamp machinery reproduces the sensitivity/criterion dissociation of Luo and Maunsell [11].

4.7 Summary: one model, the core signatures of spatial attention

Across the six analyses, the trained model reproduces several core signatures of primate spatial attention and exposes model-internal mechanisms. Behaviorally, it is a graded detector with a cued-versus-uncued benefit in VDA4; the distinct low-to-high-validity cued-threshold changes are reported in Section 6.3 (Section 4.1). Its attention deploys on a legible orient–release–re-acquire–lock schedule and differs across the VDA1 and VDA4 checkpoints, is blind to cue validity, and does not carry value in the map (Section 4.2). In signal-detection terms a single attentional clamp both lowers the decision criterion (from $c = 1.42$ to -0.60) and sets perceptual sensitivity (collapsing d' to 0.63–1.45 when attention is withdrawn, restoring it to ~ 2.0 –2.7 when returned) — the criterion account and the sensitivity account realized together in one system (Section 4.3). Misdirecting attention manufactures a confident miss (declare rate $0.68 \rightarrow 0.24$ with entropy $0.053 \rightarrow 0.035$), a mechanistic account of inattentional errors (Section 4.4). Decoding the working memory explains the behavior: value is latched at the cue and held (1.00), validity peaks at the cue (0.773) and is weak thereafter (mean post-cue 0.309), and the change is built online (0.81 presence, 0.50 location at the change frame) in a position-specific, parietal-like code (Section 4.5). And injecting priority at the monitored location causally evokes false detections ($\approx 0.03 \rightarrow 0.12$), spatially gated to where the model is attending (Section 4.6). These are the mechanisms the remaining sections carry into the value-directed (Section 5) and canonical criterion-versus-sensitivity (Section 3) paradigms.

5 Value-directed attention: the model uses spatial validity, but in an absolute-value regime

The core-detection results (Section 4) leave a pointed puzzle. On the value-directed-attention (VDA) environment the model *perceives* the predictiveness of the cue — cue validity is decodable from the working-memory state at cue onset — and yet its behavior is validity-flat: the cueing benefit does not grow as the cue becomes more reliable, and validity is only weakly recoverable after the first frame (Section 4). Taken alone this looks like a limitation. This section tests whether the pattern depends on task regime, using two separately trained manipulations. First, an environment in which spatial validity is the *only* operative cue — no competing value colour — to ask whether the capacity to use validity is present at all. Second, a relative-value manipulation to ask whether the model’s value sensitivity is context-dependent or absolute. The first is a positive association: without a competing value cue, the Validity4 checkpoint uses validity gradedly, in a Posner-like manner [4]. The second is a principled null: the model is in an absolute-value regime, consistent with the separate normative account of when value-directed attention should matter. Together they motivate, but do not prove, a cue-competition hypothesis for why the value-VDA was validity-flat, and locate the model precisely within the human and primate literature on spatial cueing and reward-directed selection.

Throughout, “the model” denotes the affine-feedback architecture. Validity4 and Baruni results are read from separately trained checkpoints, not from the canonical VDA4 weights reported in Section 4; they share the analysis harness.

5.1 Validity4: spatial validity, used gradedly

The cue is a white ring around the candidate location, and the *completeness* of that ring is the displayed validity level. This legacy Validity4 checkpoint uses a four-location target sampler that can return to the cue on the nominally invalid branch, so its realized cued-change probability

is $p + (1 - p)/4$: displayed $p = 0.25$ realizes 0.4375. It is therefore the lowest displayed-validity condition, not an uninformative or chance cue. Crucially there is no value colour: nothing on the cue signals worth, so displayed validity is the single cue dimension the model could exploit. This is a separately trained complement to the value-VDA of Section 4, where a value cue and a validity cue share the display.

Freed of the competing value signal, the model uses validity, and it uses it *gradedly*. Three converging measures, all read at a fixed change magnitude $\Delta = 26$, establish the effect.

The cueing benefit grows monotonically with validity. The spatial-cueing benefit — detection at the cued location minus detection at an uncued location — rises monotonically across displayed ring levels: it is essentially absent at the lowest displayed level and opens to a clear benefit when the cue is fully valid ($0.00 \rightarrow 0.07 \rightarrow 0.08 \rightarrow 0.12$ across validity $25\% \rightarrow 50\% \rightarrow 75\% \rightarrow 100\%$). This is a Posner-like graded cueing effect under the legacy sampler [4]: performance at the cued location improves across displayed ring levels. It also provides a cross-checkpoint contrast with the value-VDA checkpoint, where validity is present alongside value but has little measurable control over deployment (Section 4). Because the checkpoints were trained separately, the contrast shows that the architecture can learn to act on cue predictiveness when it is the only operative cue. It does not isolate cue removal within fixed weights. Validity4 and archived VDA4 use the same legacy four-location target rule, so target semantics do not confound this pair; separate training and the presence versus absence of the value cue still confound the cross-checkpoint interpretation.

Uncued detection falls as validity rises — a pattern compatible with a resource trade-off. The benefit is not merely a gain added at the cued location on top of an unchanged baseline. As the cue becomes more predictive the model *withdraws* from the uncued locations: detection at uncued locations falls from 0.94 at the lowest displayed-validity cue to 0.86 at the fully valid cue. The paired decrease is compatible with monitoring being diverted away from where the cue says a change is less likely and toward where it says a change is likely. It supports, but does not uniquely identify, reallocation of a limited attentional resource rather than a one-sided cued enhancement [4, 10]. A pure criterion shift at the cued location would leave uncued performance untouched; the falling uncued detection shows a genuine redistribution of processing, consistent with a normalization-style competition in which enhancing one location suppresses the others [5].

Cued-patch attention rises with validity. The behavioral trade-off has a direct correlate in the model’s internal attention. The fraction of attention the model places on the cued patch, α_1 (uniform allocation over the four locations = 0.25), rises with displayed validity from 0.27 — barely above uniform at the lowest displayed level — to 0.48 when the cue is fully valid. The checkpoint assigns more priority to the cued patch when the ring indicates higher reliability. This covaries with the behavioral cueing effect: a graded, validity-scaled deployment of the priority map, the model analogue of the graded top-down enhancement that spatial cues elicit in visual cortex [6, 5, 3]. That the same manipulation moves behavior (cueing benefit), the neglected alternative (uncued detection), and the internal allocation (α_1) in register supports a reallocation interpretation within this checkpoint, while the cross-checkpoint comparison remains correlational.

Taken together the three measures are a graded, Posner-like validity effect [4]: the more predictive the cue, the larger the cued benefit, the larger the uncued cost, and the more attention the model commits to the cued location. The model does not merely orient to *where* it is told to look; when validity is the operative cue it is sensitive to *how much* the cue can be trusted — the parametric dependence that was absent in the value-VDA.

5.2 The contrast with the value-VDA: a cue-competition hypothesis

The Validity4 result acquires its force from the contrast with the value-VDA of Section 4. There, on a cue ring that *also* carried a value colour, the model’s measured cueing benefit was nearly flat across displayed validity. In the value-VDA, displayed validity was most decodable from the working-memory state at cue onset (0.773 at $t = 1$), then fell near chance at the blank (0.296 at $t = 2$) and remained weak thereafter (mean $t = 2-6$: 0.309) (Section 4). The probe therefore shows strongest recoverability at the cue and only weak post-cue recoverability; it does not by itself establish active discarding.

Validity4 supplies a complementary, target-rule-matched but separately trained result. The architecture can use displayed validity when it is the only cue and can act on it gradedly (Section 5.1); both it and archived VDA4 use the legacy four-location sampler. One interpretation is that a value cue competes with validity for maintained control in the value-VDA. This cue-competition account is consistent with the reward-directed selection literature, in which reward-associated features dominate attentional priority even when other predictive information is available and even when it is suboptimal to let them do so [17, 18, 22]. The two separately trained environments therefore show that validity use is within the architecture’s repertoire but is not expressed strongly in the canonical value-VDA. They do not establish monopolization of a fixed memory resource or prove that value causes validity to be discarded; doing so would require a within-checkpoint cue intervention or matched retraining controls.

5.3 Baruni relative-value: a principled null in the absolute-value regime

The final manipulation asks a different question about value: is the model’s value sensitivity *relative* — scaled by the local context of available rewards, as in the primate relative-value paradigm — or *absolute*? The Baruni test sets a high-value and a low-value option in the same display and varies their relative worth. A relative-value system would track the *contrast* between the options; an absolute-value system would track each option’s worth on its own scale.

The model is in the *absolute*-value regime. Its response to the relative-value manipulation is flat: high-versus-low value does not modulate detection, and there is no value-directed attention — no reallocation of the priority map toward the higher-valued option as its relative worth grows. This is a *null*, and it is important to read it correctly. It is not a positive relative-value effect. It is consistent with the separate normative account’s high-sensitivity regime, in which value should be tracked absolutely, not relatively, and value-directed *attention* should be weak or absent because reallocating attention by relative value buys little when sensitivity is already high. The Baruni null is therefore a cross-paper *consistency check*: the absolute-value, weak-value-directed-attention regime identified by the normative treatment is compatible with the trained checkpoint [17, 2]. This agreement does not make the null a confirmation of the normative theory.

5.4 Summary

The value-directed-attention results resolve the puzzle left by the core detection battery and place the model in the literature. When spatial validity is the operative cue, the model uses it gradedly — a monotone, Posner-like cueing benefit ($0.00 \rightarrow 0.12$ across validity), a matching withdrawal from uncued locations ($0.94 \rightarrow 0.86$), and a validity-scaled deployment of internal attention ($\alpha_1 : 0.27 \rightarrow 0.48$) — reproducing the canonical spatial-cueing effect [4, 10]. That this graded validity use is weak in the value-VDA motivates a cue-competition hypothesis: a competing value cue may take precedence while validity drops from maintained memory (Section 4), consistent with reward-directed selection influencing attentional priority [17, 18]. And the model tracks value *absolutely*: the Baruni relative-value manipulation yields a principled null — flat detection, no value-directed attention — consistent with the high-sensitivity regime

of the separate normative account. Across separately trained checkpoints, the architecture can use spatial validity when validity is the operative cue and can display an absolute-value regime; causal value-driven crowd-out remains to be tested.

6 The value-directed attention battery: attentional signatures across set size

The reproductions above establish that a recurrent vision transformer trained with reinforcement plus temporal JEPA self-supervision (loss weight 0.5) carries the two signal-detection components of attention. We now ask how those signatures — and the spatial-cueing, sensitivity, criterion, and memory effects that accompany them — depend on the number of competing stimuli. This is the manipulation that most sharply distinguishes theories of attention: noise-limited accounts predict that the benefit of attention can *grow* with set size, because attention may protect the target signal from distractor noise [3]. We compare one canonical affine-feedback $d_{\text{mem}} = 128$ checkpoint at each of four active set sizes: one, two, four, and nine monitored Gabors (`vda1`, `vda2_2x2`, `vda4`, `vda9`). Each checkpoint has a preserved 20,000-row metrics phase ending at logged iteration 19,999. This is evidence that the logged phase completed, not a formal convergence test; resume counters may reset and metrics may be overwritten, so cumulative lifetime updates are unknown.

The comparison holds the affine-feedback mechanism and memory width in common and uses related reward and curriculum logic, but it is not an identical-model, identical-task, or pure-capacity manipulation. VDA1, the canonical VDA2_2x2 run, and VDA4 use a four-token 2×2 support with different numbers of active cells. VDA9 instead uses a 3×3 grid, nine tokens, a larger image and flattened readout, and training support that includes an additional $p = 0$ condition. The reduced-set-size and base environments also preserve different validity rules: VDA1 is degenerate; VDA2 uses `VDASetSizeEnv` and realizes displayed p exactly; archived VDA4 realizes $p + (1 - p)/4$; and archived VDA9 realizes $p + (1 - p)/9$. Thus equal displayed validity does not guarantee equal realized validity. Learned parameters are separate at every set size. Thus the four points characterize an association across these checkpoints; they do not isolate active set size from geometry, token count, readout dimension, validity semantics, or training support.

The preserved VDA16 phases are incomplete rather than budget-exhausted: their metrics stop between approximately 622 and 690 logged iterations and their newest checkpoints are at iteration 599, still near chance at the starting difficulty. They are excluded from quantitative analysis, and the available files do not reconstruct cumulative lifetime training.

Frozen legacy-artifact notice (2026-07-07). The VDA2/VDA9 change-location decode fields and the high-token VDA9 `crossattn1` clamp fields are invalid for those claims. The legacy files are preserved unchanged for provenance; corrected artifacts are pending in a versioned derived path. None of the claims below relies on those invalid fields.

Each of the four included checkpoints was subjected to the same battery of seven analyses: psychometric and chronometric functions, the attention time-course, the spatial priority map conditioned on cue value and validity, the signal-detection read-out under a graded attention clamp, the validity (cue-proportion) manipulation, and temporal decoding of cue validity from working memory. The two remaining analyses of the core detection battery — decision-uncertainty read-out and attentional microstimulation — were run on the canonical four-stimulus environment and are reported there (Sections 4.4 and 4.6); we do not restate them per set size.

6.1 Per-environment deep dives

Figures 12–15 report, for each set size, the six core panels of the battery on one page: (a) the psychometric function (detection vs. change magnitude, cued and uncued), (b) the chronometric function (reaction time vs. magnitude), (c) the attention time-course (priority on the cued vs. other patches across the seven trial frames), (d) the signal-detection criterion c as the nominal cued-location clamp is swept from suppression to enhancement, (e) the corresponding sensitivity d' , and (f) the validity effect (cued detection threshold vs. displayed cue validity). Every checkpoint is a graded detector and orients attention to the cued location. Clamp endpoints generally shift criterion in the liberal direction, but the curves are not uniformly monotone (VDA9 reverses at the high end). The across-environment *differences* are the subject of Sections 6.2–6.5.

6.2 The priority map: checkpoint-specific orienting with no consistent value steering

The per-environment panels summarize attention as a single cued-vs-other time-course. Here we read the model’s priority map *spatially*, resolving where attention falls on each cell of the attention array — the four cells of the 2×2 array for set sizes 1–4, and all nine cells of the 3×3 array for set size 9 — at each of the seven trial frames, and asking how that map is shaped by the two cue dimensions the display carries — the cue’s *value* (its colour: high, mid, or low reward, red/green/blue) and its *validity* (its ring completeness, 25–100%). Throughout, the cued location is the top-left cell, and the uniform baseline is $1/n$ for an n -cell array: 0.25 for the 2×2 set sizes and 0.11 for the 3×3 set size 9.

Orient at the cue, then release — largest at cue time in VDA9. Figure 16 shows the value- and validity-averaged attention map across the trial for each set size. The common motif is the orient-then-release dynamic already identified in the core detection analysis (Section 4.2): attention concentrates on the cued cell at the cue frame (t_1) and then relaxes as the stimulus array is processed. Across these checkpoints, the cue-time orienting response is largest in VDA9. The peak cued-cell attention at the cue frame rises from ≈ 0.26 at set size 1 — barely above the 0.25 uniform, since with a single target there is nothing to orient away from — to ≈ 0.81 at set size 9, where, against a uniform of only 0.11, the model throws almost all of its attention onto the cued cell at cue onset (0.82 of the total, versus ≈ 0.02 on each of the other eight cells) — the sharpest orienting transient of the battery. It then settles onto a *sustained* elevation of cued attention (≈ 0.31 through the array frames at set size 9, still nearly three times its uniform) that is likewise the most pronounced of the ladder. The VDA9 checkpoint therefore shows the largest cue-time concentration, but sustained attention does not form a strictly monotone four-point ladder. Given the geometry and training-support changes at VDA9, the pattern is compatible with stronger target protection under competition but does not by itself establish a noise-limited capacity law [3, 5].

Value does not consistently steer the map in these summaries. Conditioning the cued-cell attention on the cue’s *value* (Figure 17, left) reveals no value-directed steering at any set size. The difference in cued attention between the highest- and lowest-value cue is small and, critically, *inconsistent in sign* across the ladder (+0.05, −0.04, −0.01, −0.08 at set sizes 1, 2, 4, 9): a more valuable cue does not reliably attract more attention, and at the highest load the lowest-value cue if anything attracts the most. These priority-map summaries therefore provide no reliable evidence of value-based steering. This extends the value-invariance observed in the four-stimulus checkpoint (Section 4.2) to one checkpoint at each included set size. Value is tracked in memory and can affect decision variables (Section 5), while these priority-map summaries do not show reliable value-based spatial reallocation.

The VDA9 checkpoint shows the strongest validity steering. Conditioning instead on cue *validity* (Figure 17, right) shows a strong cross-checkpoint contrast. At set sizes 1, 2, and 4 the cued-cell attention is essentially flat across displayed validity (the low-to-high-validity change is -0.00 , $+0.01$, $+0.01$): the map does not track how reliable the cue is. At set size 9 it steers strongly — cued attention climbs from 0.33 at the least-valid cue to 0.47 at the most-valid cue (a change of $+0.14$), a monotone validity gradient within the VDA9 checkpoint’s priority map. This coincides with stronger validity effects at VDA9 in three readouts: *behaviour* (a 2.207° low-validity minus high-validity *cued-threshold* improvement, not a cued-versus-uncued spatial benefit; Section 6.3), *working memory* (maintained validity decoding, Section 6.5), and now the *priority map*. Because this is a cross-checkpoint association with a concurrent grid and training-support change, it does not establish that load alone caused this conjunction. It also accords with the isolated-validity environment (Section 5.1), where removing the competing value colour lets validity steer the map ($\alpha_c : 0.27 \rightarrow 0.48$) even at moderate load: validity can steer the priority map under at least two separately trained conditions. The conjunction is compatible with the hypothesis that cue dimensions compete for retention or control (Section 5.2), but these data do not show that value monopolizes a fixed resource or that validity re-enters at a causal load threshold.

6.3 Validity-threshold changes differ across checkpoints

The cued detection threshold changes little across displayed cue validity at set size 1 and more strongly in the VDA9 checkpoint (Figure 18). Defined specifically as *low-validity minus high-validity cued threshold*, the four improvements are 0.148° , 0.193° , 0.979° , and 2.207° for VDA1, VDA2, VDA4, and VDA9. The 2.207° quantity is therefore not a spatial cued-versus-uncued benefit. The ordered increase is consistent with greater use of a predictive cue as competition increases [3], but the single-checkpoint design and the VDA9 geometry, readout, validity-semantic, and training-support differences prevent interpreting it as a pure capacity law. It also qualifies an apparent paradox in the single-environment analysis (Section 4), where validity looked “inert”: the four-stimulus checkpoint has a modest threshold association that is easier to see in the cross-checkpoint summary.

6.4 Clamp-induced criterion and sensitivity ranges differ across checkpoints

Sweeping the nominal cued-location clamp and reading out signal-detection quantities at each set size (Figure 19) yields two patterns that must be kept separate from the validity-threshold improvements above. For each quantity we report its maximum minus minimum over the clamp sweep. The sensitivity ranges are 1.982, 1.647, 1.131, and 0.367 for VDA1, VDA2, VDA4, and VDA9. The criterion ranges are 1.020, 1.381, 1.622, and 1.048. Thus sensitivity range decreases across these four checkpoints, whereas criterion range is non-monotone and largest at VDA4. The small VDA9 d' range does not demonstrate physiological saturation or the absence of a sensitivity contribution; it is an observed clamp range from one checkpoint. Moreover, the VDA9 criterion curve reverses at the high-clamp end, so a blanket monotonic-criterion description is inaccurate.

Because d' range, criterion range, and low-to-high-validity threshold improvement are different outcomes, we do not combine them into one scalar “attentional benefit” or capacity law. Their differing profiles instead motivate a controlled replication that holds grid, tokens, readout, validity semantics, and training support fixed. The present results establish descriptive, checkpoint-specific load associations and a causal within-checkpoint clamp response, not a single mechanism that changes monotonically with set size.

6.5 Validity decoding from working memory differs across checkpoints

Finally we ask whether the *maintenance* of cue validity in working memory — not just its behavioural expression — depends on set size. Decoding the displayed cue validity from the recurrent memory state at each frame (balanced accuracy, held-out four-way logistic probe; chance = 0.25) tests whether the model perceives and holds the cue’s reliability (Figure 20). The result is strongest at VDA9 but is not a monotone ladder at the intermediate checkpoints. At set size 1, validity is barely encoded even at the cue frame (peak 0.35) and is at chance thereafter — the model does not hold a cue reliability that cannot help it. At set sizes 2 and 4, validity is strongly perceived at the cue (peak 0.85 and 0.77) but becomes only weakly linearly recoverable within a frame or two (mean post-cue accuracy 0.35 and 0.31, near chance) — the cue-locked pattern reported for the single four-stimulus environment (Section 4). In VDA9, however, the validity-decoding point estimate is 1.00 at the cue and is *maintained across the entire trial* (mean post-cue accuracy 0.70, still 0.51 at the final frame). The mean post-cue balanced accuracies for VDA1, VDA2, VDA4, and VDA9 are approximately 0.247, 0.350, 0.309, and 0.701: VDA9 is clearly highest, while VDA2 and VDA4 do not order monotonically.

The contrast with the *value* signal on the same cue is stark and completes the account. Decoding the cue’s value (its colour) from the identical memory state yields a approximately 1.00 at the cue frame and subsequent sampled frames in every included checkpoint (Figure 20, dashed). Value is thus linearly recoverable at ceiling in these evaluations; validity is encoded well but released within a frame or two unless the load is high. The two cue dimensions therefore receive different treatment in these checkpoints: value is retained at ceiling, whereas validity is retained most strongly in VDA9. This pattern is compatible with the proposed cue-competition account (Section 5.2) and accompanies the VDA9 validity-steered priority map (Section 6.2); it does not establish value monopolization or a fixed representational share.

This is a memory-level counterpart of the behavioural VDA9 contrast (Section 6.3): the VDA9 checkpoint retains cue reliability more strongly than the other canonical checkpoints. Whether active load makes that reliability worth holding is a hypothesis for a fixed-geometry, replicated intervention, not a functional result established by the present cross-checkpoint comparison. The pattern is nevertheless consistent with task-relevance-gated encoding described for prefrontal working memory [23, 13].

7 Working-memory width and controlled detection

In humans, working-memory capacity predicts the ability to exert controlled attention — to suppress a pre-potent orienting response and deploy attention where the task requires [15]. The model offers a related, but not identical, intervention: changing the width of its spatial recurrent state (d_{mem}). Width is an architectural parameter, not a direct measurement of human working-memory span, so any capacity interpretation must be earned by a matched behavioral analysis.

Preserved $d_{\text{mem}} = 128$ and 256 checkpoints now exist for VDA1, VDA2, VDA4, and VDA9. Their existence corrects the earlier status that the larger-width run was unavailable. Training summaries do not show a simple monotone width benefit: affine-feedback checkpoints learn at both widths, while the $d_{\text{mem}} = 256$ VDA9 cross-attention run is unstable late in training. Training accuracy alone does not establish a controlled-attention effect.

Status and planned comparison. A matched held-out battery comparing $d_{\text{mem}} = 128$ with 256 — psychometrics, invalid-cue behavior, temporal decoding, and the causal attention clamp, with checkpoint identities fixed — has not yet been completed. We therefore report no capacity law and no direction of a width effect. The preserved checkpoints make that analysis possible; until it is run, memory width remains an intervention whose behavioral meaning is unresolved.

8 Attention-based rehearsal remains a hypothesis

A long-standing hypothesis holds that spatial working memory is maintained by covertly shifting attention toward the memorized location — the spatial analogue of articulatory rehearsal — so that the usual consequences of spatial attention should be measurable at the remembered location during a delay [24, 13].

The present model provides observations compatible with this hypothesis but not a rehearsal result. It later re-acquires cue-directed attention after the blank, and some checkpoints show elevated cued-location activity during a blank frame. No cue-location decoder was fitted across the delay, so these are indirect signatures compatible with maintenance rather than a decoded maintained-location result. In the canonical affine VDA4 trace, however, the measured cued attention share falls near the uniform baseline during the blank ($\alpha_c \approx 0.22$ versus 0.25). Neither pattern by itself shows that attention is the load-bearing maintenance mechanism.

A delay-probe environment has been implemented, but we found no trained probe checkpoint, memorized-location enhancement result, or causal delay-clamp analysis to report. The decisive tests remain prospective: compare probe processing at cued and uncued locations during the delay, then suppress cued-location attention during that period and test whether any probe advantage or later detection benefit is reduced. Until those measurements exist, attention-based rehearsal is a hypothesis rather than an empirical claim of this manuscript.

9 Discussion

An affine-feedback recurrent vision-transformer architecture with a spatial working-memory state, trained with reinforcement plus temporal JEPA self-supervision (loss weight 0.5) to detect a change among cued Gabor stimuli, reproduces several functional signatures of primate visual attention across a core checkpoint and separately trained task variants and — through the attention-clamp battery — takes several of its long-standing open questions a step past what behaviour alone can resolve. The model orients to a cue, relaxes near baseline during the blank and later re-acquires cue-directed attention, gates its own detection sensitivity, shifts its decision criterion, and holds a linearly decodable, position-specific memory signal after the change. No cue-location decoder was fitted across the delay. These are observed signatures of checkpoints optimized jointly by RL and a JEPA auxiliary loss (weight 0.5), not hand-specified attention targets. We organise what follows around three claims the results support — that the model directly operationalizes criterion and sensitivity effects plus spatial gating, that it exposes interactions between attention and working memory, and that the joint architecture is usefully read against a counterpart in the parietal priority map — and then delimit, plainly, where the account is incomplete.

Directly measured criterion, sensitivity, and spatial gating

For twenty-five years the field has carried three competing explanations of how attention improves perception: attention enhances the signal (gain), it reduces external noise and selects the target more efficiently, or it acts at the decision stage by shifting the observer’s criterion and reducing spatial uncertainty [3]. The three “are not mutually exclusive; rather, all three are likely to contribute” [3]. The present battery does not estimate how much each contributes. Instead, it directly measures criterion and sensitivity under the same attention clamp and tests whether an injection effect is spatially gated. It contains no dedicated external-noise manipulation, so efficient external-noise exclusion remains compatible with the results rather than quantitatively isolated.

The spatial-gating result is direct: attention added to the monitored patch raises detection, and injecting priority onto a patch on no-change trials conjures a false detection — the false-alarm rate rises roughly fourfold, from about 0.03 to about 0.12, as injection strength grows

(Section 4). This is an intervention effect in the decision pathway. It is spatially gated: injecting priority at the monitored location evokes the false percept, whereas injecting at an unmonitored patch does essentially nothing (a flat $\sim 0.05\text{--}0.10$); the model reads out the location it is already selecting, not priority in the abstract (Section 4). A cued-versus-uncued benefit is present in VDA4, whereas VDA1 has no uncued active item and therefore does not define the same spatial contrast. Separately, the low-to-high-validity *cued-threshold* changes are 0.148° , 0.193° , 0.979° , and 2.207° across VDA1/2/4/9. The 2.207° endpoint is not a spatial cued-versus-uncued benefit. These associations are compatible with a noise-limited selector [3, 10], but VDA9 also changes grid, token count, readout dimension, validity semantics, and training support, and they are not an external-noise test. *Decision criterion*: in the canonical VDA4 checkpoint, sweeping the nominal clamp from suppression to enhancement moves the sampled signal-detection criteria in the liberal direction, $c : 1.42 \rightarrow -0.60$ (Section 4) — the textbook attentional criterion shift [10, 4], now produced by an analysis intervention rather than a payoff instruction. This monotonic description is local to VDA4; VDA9 reverses at the high-clamp end.

The criterion and sensitivity endpoints are both operationalized on the *same* clamp. The criterion moves smoothly across the VDA4 sweep; sensitivity is load-bearing on attention — suppressing attention collapses d' (to $0.63/1.45$ in the suppression regime) while natural and enhanced attention restore it (to $\sim 2.0\text{--}2.7$) (Section 4). These are two directly measured responses to the intervention, not a quantitative three-way contribution decomposition. They parallel the distinction the normalization account of gain predicts at the neural level [5, 6, 7] and which psychophysics can only reach indirectly. The same contrast appears in a preliminary cued analogue of the Luo–Maunsell paradigm: in separately trained reward-regime checkpoints, concentrating value at one location moves sensitivity ($\Delta d' = 1.09$ at $\Delta = 18^\circ$, with a low false-alarm rate of 0.02) while a global reward that spares no location moves the criterion instead (false-alarm rate 0.18 , sensitivity change of only 0.59) (Section 3) [11]. That the two levers separate under reward structure, and that related quantities separate under a direct attention clamp in the base task, is encouraging. It is not yet a faithful cue-free, shared-base reproduction, and the between-checkpoint comparison should not be read as a within-model double dissociation.

One further, novel read-out sharpens the criterion story into a failure mode. When attention is misdirected onto an empty monitored patch while a real change occurs elsewhere, the model misses the change — the declare rate falls from 0.68 to 0.24 — and does so while becoming *more* confident, its policy entropy dropping from 0.053 to 0.035 (Section 4). Misallocated attention does not merely lower performance; it manufactures confident perceptual errors, because the detector reports “no change” precisely because it was pointed at the wrong place. This is a mechanistic account, in one system, of how an attentional misallocation — the sort a misleading cue induces — can produce not just misses but assured misses.

One architecture links attention to working memory

The second claim is that attention and working memory interact within one recurrent architecture. The current observations establish coupling, not identity: the memory state shapes attention, attention gates sensitivity, and linear probes recover selected task variables from memory at sampled frames.

The model *reads value*: cue colour, which signals the worth of a correct detection, is latched into the memory state at cue onset and held perfectly — decoding balanced accuracy 1.00 from the first frame through to the change (Section 4). It *deploys spatially*: attention orients to the cued patch at cue onset, well above the uniform baseline, so that the memory of *where* to look is expressed as a priority map over the visual field (Section 4). It *gates its own sensitivity*: as shown above, detection d' rises and falls with the attention placed on the monitored location, so the deployed attention is not a spectator but the quantity that sets how well the change can be seen. Its blank-period state is compatible with a rehearsal hypothesis, but does not establish one. In the canonical affine VDA4 trace, cued attention falls near the uniform baseline during the

blank (0.22 versus 0.25), and no delay-probe or causal delay-clamp result is reported (Section 8). Whether covert spatial attention is the load-bearing maintenance mechanism therefore remains a prospective test [23, 25]. The later cue-directed re-acquisition is compatible with an intervening spatial trace, but no cue-location decoder was fitted across the blank.

What distinguishes reading value from perceiving predictiveness is instructive. Cue *validity* — how reliably the cue predicts the change location — is linearly decodable at the cue but falls toward chance within the next sampled steps (Section 4). That weak maintenance accompanies the nearly flat validity effect in the base task, but decoding alone does not prove that loss of linear recoverability is its unique cause. The separately trained Validity4 task shows that the architecture can use validity when it is the only cue on offer. There the checkpoint uses it gradedly: the cueing benefit grows monotonically with displayed validity ($0.00 \rightarrow 0.07 \rightarrow 0.08 \rightarrow 0.12$ from 25% to 100%), uncued detection falls as the cue sharpens ($0.94 \rightarrow 0.86$), and attention on the cued patch rises with validity ($\alpha_c : 0.27 \rightarrow 0.48$ against a uniform 0.25) (Section 5). A Posner-style validity effect [4] is thus within the architecture’s repertoire. Under the legacy Validity4 sampler, displayed $p = 0.25$ realizes 0.4375 cued-change probability; archived VDA4 uses the same four-location rule. Target semantics therefore do not confound this pair, although the separately trained checkpoints and the presence versus absence of a value cue do. Their contrast is compatible with cue competition but does not establish that value monopolizes a fixed resource or causes validity to be discarded. In these checkpoints, value remains linearly recoverable from cue onset, whereas validity maintenance depends on task and checkpoint.

A biological anchor: the priority map that binds position memory to detection threshold

These behaviours motivate a functional analogy to a parietal priority map. After the change, the model combines a position-specific event-location signal with an attentional modulation of detection threshold (the clamp moves d' and c). The current analyses do not directly quantify a trial-level coupling between representational strength and threshold. They nevertheless place both measurable functions within one recurrent architecture, paralleling accounts of area LIP and allied parietal circuits in which spatial representations track task-relevant locations and bias visual processing there [16].

We make this correspondence as an *honest analogy, not an isomorphism*. The model’s memory is a recurrent state shaped by joint RL and JEPA optimization (auxiliary weight 0.5); LIP is a population of parietal neurons embedded in a cortical–subcortical circuit, and no element of the model corresponds to a specific cell class, layer, or projection. What transfers is the proposed computation — event-location coding and attentional threshold modulation within one architecture — not anatomy or an established single-signal identity. A direct relationship between code strength and threshold remains to be tested.

Completed battery and open tests

The affine $d_{\text{mem}} = 128$ VDA1/2/4/9 battery is reported in full in Section 6; it is not forthcoming. Each point uses one canonical checkpoint with one 20,000-row logged phase. The battery shows an ordered increase in low-to-high-validity cued-threshold change, but sustained attention, criterion range, and validity maintenance do not form a single monotone law. VDA16 is an interrupted run ending near chance at checkpoint 599, not a completed endpoint or an exhausted-budget failure.

Two interventions remain open. Preserved $d_{\text{mem}} = 128$ and 256 checkpoints exist, but the matched held-out psychometric, decoding, and clamp battery has not been completed; no width or capacity law is reported (Section 7). The delay-probe environment is implemented, but no trained probe checkpoint, memorized-location advantage, or causal delay-clamp result is reported. Attention-based rehearsal therefore remains a hypothesis (Section 8).

Scope and limitations

Several boundaries delimit these statements. The strongest is a training outcome, not an interpretive caveat: not every attentional paradigm we attempted was learnable by the model as posed. An attend-here/ignore-here filtering task did not train — the policy collapsed into always withholding, never declaring a change, despite an environment whose reward structure is achievable in principle. This failed task is not reported as a quantitative Results section; its run-level trace is identified in the Appendix. The collapse is a learning failure of a particular curriculum, not evidence that the mechanism is absent, but it means we cannot report the corresponding selective-attention result, and we say so rather than presenting a partial run as a finding. Relatedly, the Validity4 checkpoint uses cue validity when validity is the operative cue, whereas VDA4 weakly maintains validity when a competing value cue is present (Sections 4 and 5). Because these are separately trained checkpoints, the contrast is a task association rather than a within-weight test of cue competition and should not be read as a general Posner effect.

The set-size characterization is complete for the four included checkpoints but is single-checkpoint and confounded by task/model differences. The matched width analysis and delay-probe rehearsal result are not in hand, and nothing in the discussion should be read as asserting their outcomes. The Baruni relative-value test is a principled null rather than a positive result: the model’s value sensitivity is absolute, not relative to context (Section 5), which is consistent with the regime the model occupies but is a null nonetheless [12]. Finally, the model family is trained on a narrow family of change-detection tasks. The behaviours it reproduces are broad, but the reproductions are functional signatures — psychometric, signal-detection, and decoding profiles — and the biological correspondences are analogies at the level of computation. The contribution is that an instrumented jointly RL-JEPA-trained architecture (auxiliary weight 0.5), dissected with causal clamps and linear probes, separates several co-present attentional signatures while exposing a working-memory state whose relation to rehearsal remains a prospective causal test.

10 Methods

This section specifies the model, the environments, the training procedure, and the analysis harnesses in full. Section 2 introduced the architecture conceptually and stated the load-bearing equations; here we give the pixel-level task timeline, the layer-by-layer network, the exact objective and hyperparameters, and the causal and read-out analyses that generate the results of Sections 4–3. The model is a single recurrent vision transformer with element-wise affine memory→vision feedback (Equation 1), trained end to end with reinforcement learning and a temporal JEPA auxiliary loss (weight 0.5). The core causal battery uses one canonical checkpoint; set-size and paradigm sections use separately trained checkpoints and state that fact locally. Reported evaluation quantities are single-checkpoint estimates unless a distribution over trials is stated; they do not estimate between-training-run uncertainty.

10.1 The change-detection environment

The task is a covert cued orientation-change-detection paradigm, rendered as a short sequence of 50×50 RGB images with pixel intensities in $[-1, 1]$, adapted from the cued-attention paradigms of the primate and psychophysical literature [4, 10, 3]. The core environment places $N = 4$ Gabor stimuli on a 2×2 grid, one per 25×25 quadrant, indexed row-major so that quadrant, cue, and change indices coincide (top-left = location 1, bottom-right = location 4). Set size is a manipulated variable (Section 6), but the preserved comparison is not a pure active-item intervention. The canonical `vda1`, `vda2_2x2`, and `vda4` checkpoints all use a four-token 2×2 support with one, two, or four active cells. The `vda9` checkpoint uses a 3×3 grid, nine active

tokens, a 75×75 image, and a correspondingly larger flattened decision readout; its training support also includes a displayed-validity $p = 0$ condition.

The archived checkpoints preserve three historical validity rules. VDA1 is degenerate because the cue is the only active target. VDA2 was generated by `VDASetSizeEnv`, whose invalid branch samples only non-cued active items, so its displayed and realized validity agree. Archived VDA4 and VDA9 instead use the legacy sampler, whose invalid branch is uniform over all four or nine locations, including the cue; their realized cued-change probabilities are respectively $p + (1 - p)/4$ and $p + (1 - p)/9$. The separately trained Validity4 checkpoint uses the same legacy four-location rule as archived VDA4, so target semantics do not confound that pair. Exact-validity code dated 2026-07-11 applies only to future corrected runs; no completed corrected derived artifact exists. The 4×4 `vda16` runs are interrupted pilot runs and are not included in the quantitative ladder.

Trial timeline. A trial is $T = 7$ logical frames indexed $t = 0, \dots, 6$ (Figure 21):

- $t = 0$: blank (all-zero) frame.
- $t = 1$: the *cue*, a coloured disc-and-ring rendered in the quadrant of the cued location (a single location drawn uniformly among the admissible cue positions); all other quadrants are blank.
- $t = 2$: blank frame (the delay).
- $t = 3, 4, 5, 6$: the four oriented Gabor patches, one per location, present and held through the end of the trial.

The change, when present, occurs at a *fixed* frame, $t = 5$: the minimum and maximum change frames are both set to 5, so the model faces a temporally predictable change onset and the change is visible on frames 5 and 6. On 50% of trials (drawn independently each trial) a change is present and exactly one patch’s orientation steps by Δ from $t = 5$ onward; on the remaining 50% no change occurs.

Gabor stimuli and orientation noise. Each patch is a Gabor grating (Gaussian envelope $\sigma = 0.5$ on the $[-1, 1]$ patch square, wavelength $\Lambda = 0.3$, aspect ratio 1, phase 0), masked to a central disc, with low-amplitude additive pixel noise (uniform on $[-0.11, 0.11]$). Each patch is assigned an independent base orientation drawn uniformly on $[0^\circ, 360^\circ)$ at trial start, and on *every* stimulus frame an independent orientation jitter is added, $\eta \sim \sigma_{\text{ori}} \mathcal{N}(0, 1)$ with $\sigma_{\text{ori}} = 5^\circ$. The per-frame jitter is the internal noise the model must integrate through: a single-frame reader cannot separate a genuine orientation step from the jitter, so detection requires accumulating evidence across the held stimulus frames. The change is a sustained orientation offset Δ added to the changed patch’s orientation from $t = 5$ on (superimposed on the ongoing jitter), with the signed magnitude drawn per trial as $\Delta \sim \mathcal{U}(-\theta, \theta)$; θ is the curriculum-controlled difficulty level (Section 10.3).

Cue: location, value, and validity. The cue at $t = 1$ is a filled central disc (radius 0.32 of the patch side) plus a surrounding partial ring (inner/outer radii 0.40/0.48 of the patch side). Two graded properties are carried on this single cue:

- **Validity** is encoded by the *ring completeness*, a proportion p drawn uniformly from $\{0.25, 0.5, 0.75, 1.0\}$ (the fraction of the ring’s circumference that is drawn). For the archived runs, VDA2 realizes p exactly, VDA4 and Validity4 realize $p + (1 - p)/4$, VDA9 realizes $p + (1 - p)/9$, and VDA1 is degenerate. We report the displayed ring proportion

and identify the historical rule in cross-checkpoint comparisons. The 2026-07-11 exact-validity implementation samples invalid changes only from non-cued active locations but has no completed corrected run artifact.

- **Value** is encoded by the *cue colour*. In the value-directed environment the colour signals the reward for correctly detecting a change at the cued location: red = 5, green = 3, blue = 1, drawn uniformly per trial. When value is not manipulated the cue is white and every correct response earns unit reward.

Actions and reward. The action space is binary at every frame: *wait* (0) or *declare-change* (1). Declaring on or after the change frame on a genuine reportable-change trial earns the trial’s reward magnitude (the cue-colour value in the value environment, else 1); declaring before the change, or on a no-change trial, is a false alarm and ends the trial with zero reward. Waiting through a no-change trial to the final frame earns the reward magnitude as a correct rejection. Because both hits *and* correct rejections are rewarded, an always-wait policy is not trivially optimal — it forgoes all hit reward on the 50% of change trials — and the model must learn to discriminate a real orientation step from the per-frame jitter. A trial that is neither declared nor a correct rejection by the final frame terminates without further reward. This reward structure corresponds to the value-coupled regime in which correct responses (both hits and correct rejections) are paid the value magnitude of the cued location.

Variants. The battery of Sections 5 and 3 realises established primate paradigms by overriding only the reward, target, or response rules of the core environment, leaving the timeline, rendering, and change logic untouched:

- **Value-directed (core).** Coloured value cue + validity ring, as above; the setting for the core-detection results (Section 4).
- **Validity (pure spatial cueing).** A separately trained Validity4 checkpoint with no value colour; a white cue whose ring completeness carries validity only, with the cue restricted to the two diagonal corners. This checkpoint retains the legacy four-location target sampler above, the same historical rule used by archived VDA4. It removes the value colour but remains a separately trained rather than within-weight VDA4 control (Section 5).
- **Reward-structure sessions (Luo–Maunsell).** Two reward regimes on the same stimuli — a *sensitivity* session in which the reward for a correct detection is concentrated at one (cued) location (high reward = 5 there, 1 elsewhere), and a *criterion* session in which every hit and correct rejection is paid a fixed ratio-controlled reward independent of location — together with a response window of 2 frames after change onset, beyond which an undeclared change expires as a miss. This dissociates a sensitivity manipulation from a criterion manipulation on matched stimuli [11] (Section 3).
- **Relative-value (Baruni).** A value-directed session used to test whether the model’s value sensitivity is relative (context-scaled) or absolute (Section 5).
- **Delay-probe (rehearsal; implemented, not analysed).** The implemented environment inserts a transient probe Gabor in the otherwise-blank delay frame ($t = 2$) at either the cued (memorised) location or a random uncued location, carrying no reward, to test for enhanced processing at the memorised location during the delay [24] (Section 8).
- **Attend-here/ignore-here filtering (failed training attempt).** Only a change at the cued location is reportable; a foil change should be ignored. The attempted checkpoints collapsed to always waiting, so this variant contributes no quantitative result. The preserved run-level summary is traced in the Appendix.

All environments share the $T = 7$ timeline, the fixed $t = 5$ change, the 50% change probability, the $\sigma_{\text{ori}} = 5^\circ$ per-frame jitter, the $\Delta \sim \mathcal{U}(-\theta, \theta)$ change magnitude, and the curriculum below.

10.2 Network

The network chains a convolutional front-end, a recurrent self-attention encoder with element-wise affine feedback, a per-location spatial working memory, and a distributional actor-critic, with an auxiliary temporal self-distillation head; the components were motivated in Section 2. We give the exact layer specification here.

Convolutional front-end. In the core four-location checkpoint, each 50×50 RGB frame is split into the four 25×25 quadrant patches in row-major order. A single shared squeeze-and-excitation residual network encodes each patch: a 3×3 convolutional stem ($3 \rightarrow 32$ channels) followed by three squeeze-and-excitation residual stages that downsample and grow channels ($32 \rightarrow 64$, $25 \rightarrow 13$; $64 \rightarrow 128$, $13 \rightarrow 7$; $128 \rightarrow 128$, $7 \rightarrow 7$, added depth without further downsampling), a global average pool to a 128-vector, and a root-mean-square normalisation. Each residual block is convolution-groupnorm-SiLU $\rightarrow$ convolution-groupnorm $\rightarrow$ squeeze-and-excitation channel gate $\rightarrow$ additive (projected) residual $\rightarrow$ SiLU. Group normalisation (rather than batch normalisation) keeps the front-end train/eval-identical and stable at the batch size of one imposed by the online recurrent rollout; root-mean-square normalisation (rather than a mean-subtracting layer normalisation) preserves the uniform colour (DC) offset that carries the value cue. The 128-dimensional patch embedding $\hat{o}_i$ is concatenated with a one-hot location code (N -dimensional) and a one-hot temporal code (8 slots, indexed by the logical frame), giving a 140-dimensional token per patch and a token matrix $X \in \mathbb{R}^{N \times 140}$ per frame. The front-end carries ≈ 0.59 million parameters and is trained end to end from task reward, with no pretraining and no reconstruction loss.

Recurrent self-attention with element-wise affine feedback. The tokens pass through a single recurrent self-attention layer. Before attending, the tokens are affinely modulated by the previous-step working-memory state $H \in \mathbb{R}^{N \times 128}$: a control vector $b = \tanh(\text{bottleneck}(H))$ of width 64 is projected to a per-element scale $\gamma(H)$ and shift $\beta(H)$ of the token shape, and the tokens are transformed by $X' = \gamma(H) \odot X + \beta(H)$ (Equation 1). The scale bias is initialised to one and the shift to zero, so $X' = X$ at initialisation and the layer begins as ordinary self-attention; the memory’s gain-and-bias field is learned. The modulated tokens undergo standard scaled dot-product self-attention over the N locations, $Q = X'W_Q$, $K = X'W_K$, $V = X'W_V$, $A = \text{softmax}(QK^\top/\sqrt{d})$, $Z = X + AV$ (Equation 2), with the raw tokens X retained as a residual so the current frame always reaches memory regardless of the attention weights. The attention matrix $A \in \mathbb{R}^{N \times N}$ is the model’s internal allocation of processing across space; its row-marginal α_i is the attention share on location i , and is the quantity read out and manipulated in the analyses below. The pre-softmax attention logits accept an additive per-key clamp used only in analysis (Section 10.4.1); it is inert during training.

Spatial working memory. The contextualised tokens Z are integrated across the trial by an extended-LSTM (xLSTM) cell applied per location with shared parameters but a private state per patch, producing a 128-dimensional memory vector H_i per location. Writing the stabilised update, the input, forget, output, and cell-input pre-activations are $\tilde{I} = ZW_i + HR_i$, $\tilde{F} = ZW_f + HR_f$, $\tilde{O} = ZW_o + HR_o$, $\tilde{U} = ZW_u + HR_z$ (input maps $W \in \mathbb{R}^{140 \times 128}$ biased, recurrent maps $R \in \mathbb{R}^{128 \times 128}$ bias-free); the log-domain stabiliser is $M = \max(\tilde{F} + M_{\text{prev}}, \tilde{I})$ with $I = \exp(\tilde{I} - M)$ and $F = \exp(\tilde{F} + M_{\text{prev}} - M)$; and the state updates are $O = \sigma(\tilde{O})$, $U = \tanh(\tilde{U})$, $N = F \odot N_{\text{prev}} + I$, $C = C_{\text{prev}} \odot F + U \odot I$, $H = O \odot (C/N)$, with the state (H, C, N, M) per patch initialised to zero. The stabilised exponential gates give a controllable

retention time-constant, so the memory can encode cue information across the blank delay and accumulate weak orientation evidence across the stimulus frames. The memory state at frame t is the H that drives the affine feedback (Equation 1) at frame $t + 1$, closing the perception–memory loop.

Distributional actor–critic. The N per-location memory vectors are flattened into a single $N \times 128$ -dimensional readout (512 dimensions in the core four-token checkpoint and 1152 in `vda9`); the spatial structure is discarded at the decision stage. Two heads that share no parameters read it, each a three-layer feed-forward trunk (width 256, ELU nonlinearities). The *actor* maps the readout to logits over the two actions and is biased at initialisation toward *wait* (bias $[0.0, -1.5]$) so the model does not begin by declaring on every frame. The *critic* is a distributional quantile head: it outputs 5 return quantiles per action (quantile-regression value head), and the state value is recovered as the action-probability-weighted mean of the per-action quantile means. The decision is made online, frame by frame, which gives the model a reaction time and hence a chronometric function.

Auxiliary temporal self-distillation. A joint-embedding predictive self-distillation head (a temporal V-JEPA objective with DINO-style centering) reads the memory cell at each location and predicts, for the same location one frame ahead, the soft prototype assignment of an exponential-moving-average teacher copy of the network. The head emits, per location, 4 independent softmax distributions over 256 prototypes each. It has no direct forward connection to the actor, critic, attention logits, or next recurrent state. Its loss nevertheless backpropagates through shared representations and can therefore influence the learned features and behaviour.

The core model uses a per-location memory width of $d_{\text{mem}} = 128$ throughout; the working-memory-capacity manipulation of Section 7 varies d_{mem} (128 versus 256) with all else fixed.

10.3 Reinforcement-learning objective

The total training objective combines an off-policy maximum-a-posteriori actor–critic objective from sparse task reward with the temporal JEPA self-distillation loss (weight 0.5). The reinforcement component uses a distributional critic, prioritised episodic replay, and an exponential-moving-average target.

Actor. The actor is updated by a maximum-a-posteriori policy-optimisation step: an expectation step reweights actions by their advantage under the exponential-moving-average reference policy at temperature $\eta = 0.1$, followed by a weighted maximisation with a behavioural-cloning term (behavioural-cloning weight 0.1; 0 recovers pure policy optimisation, 1 pure cloning) and an entropy regulariser (coefficient 0.01). The reference policy in the expectation step is the target network, not the online network.

Critic. The critic is trained by a quantile-regression temporal-difference loss over its 5 return quantiles, with quantile-Huber threshold $\kappa = 1.0$ and value-loss weight 0.5. The bootstrap target is computed from the exponential-moving-average target network. The discount is $\gamma = 0.95$.

Replay and target. Learning uses prioritised *episodic* replay: complete trials are stored in a first-in-first-out buffer (capacity 1000 episodes) and sampled by priority (priority exponent 0.6, importance-sampling exponent annealed $0.4 \rightarrow 1.0$, priority clip 50), with 4 replayed episodes alongside the fresh on-policy episodes each iteration, so informative whole trials are revisited in proportion to their surprise. The target network is an exponential moving average of the online network (decay 0.995, applied to parameters and buffers each iteration), and supplies both the critic bootstrap target and the actor’s reference policy, replacing a hard periodic copy.

JEPA teacher. The self-distillation teacher is a separate exponential-moving-average copy of the network (parameters only, decay 0.996, decoupled from the reinforcement-learning target), and the student at frame t predicts the teacher’s soft assignment at frame $t + 1$. DINO-style centering uses centre-momentum 0.9; the student temperature is 0.1 and the teacher temperature is warmed from 0.04 to 0.07 over the first 300 iterations. The self-distillation loss enters the total objective with weight 0.5.

Optimisation. Parameters are optimised with a learning rate of 3×10^{-4} and gradient-norm clipping at 0.5. Each iteration collects 8 episodes, with a burn-in of 20 iterations before replay-based updates begin. Each canonical affine $d_{\text{mem}} = 128$ VDA1/VDA2_2x2/VDA4/VDA9 run has one preserved metrics file with 20,000 rows ending at logged iteration 19,999 and one canonical final checkpoint. This documents a completed logged phase, not formal convergence. Resume jobs can reset iteration counters and overwrite `metrics.csv`, so cumulative lifetime updates cannot be reconstructed from those files.

Difficulty curriculum. Difficulty is set by the change-magnitude bound θ in $\Delta \sim \mathcal{U}(-\theta, \theta)$, and is shrunk by a performance curriculum so the model is always trained near the edge of its competence. θ starts at 65° ; correctness is averaged over *non-overlapping* windows of 1000 trials, and whenever a completed window is solved at $\geq 85\%$ correct, θ is reduced by 3° , floored at 8° . Shrinking θ makes changes smaller and thus harder to discriminate against the fixed 5° per-frame jitter, so the model is driven toward finer sensitivity as it improves. Because the change is signed and drawn symmetrically, at any θ the trial pool spans the full range of change magnitudes up to θ , and psychometric functions are constructed by binning trials on the realised $|\Delta|$ rather than on θ .

10.4 Analysis harnesses

Each analysis is run off-policy on the frozen checkpoint named by its result section, by re-encoding whole trial sequences and reading out (or intervening on) the internal quantities. Unless stated, an analysis condition is estimated over a fixed batch of freshly sampled trials at the model’s curriculum-hardened operating point, and the change magnitude is set to a diagnostic value (e.g. $\Delta = 18^\circ$ or $\Delta = 28^\circ$) or swept to build a psychometric function.

10.4.1 Attention clamp: microstimulation and graded biasing

The single causal lever is an additive bias on the pre-softmax attention logits of a chosen key location, applied at inference only. Writing a bias β_j on the logits of key location j shifts the softmax allocation A toward ($\beta_j > 0$) or away from ($\beta_j < 0$) that location, without altering the trained weights. Figures parameterise the intervention by a nominal clamp value, denoted α_{nom} , which is mapped to this additive logit bias. It is an intervention setting, not a measured post-softmax attention share; realized attention is read separately from A . We use the lever in three modes, all mirroring interventional primate paradigms in which a spatial priority signal is injected or removed:

- **Graded biasing.** Sweeping the nominal clamp from suppress to enhance, mapping it to a negative or positive β on the cued key, and reading out signal-detection quantities and behaviour against that nominal setting (Section 10.4.2, Section 4.3). This is a within-model dose-response on the attentional allocation.
- **Hard override.** Clamping the allocation onto a specified location ($\alpha \rightarrow 1$ there) on trials whose change is elsewhere, to test whether forcing attention onto (or off) a location changes detection — the analogue of enhancing or inactivating a spatial priority representation [10].

- **Microstimulation.** Injecting attention (positive bias) onto a location on *no-change* trials, to test whether surplus priority conjures a false detection, and whether the evoked percept is spatially gated to the monitored location (Section 4.6). The injection strength is swept and the induced false-alarm rate recorded at the cued versus an uncued site, the causal converse of the criterion read-out.

10.4.2 Signal-detection read-out

Behaviour is scored in signal-detection terms on matched change and no-change trials. A *hit* is a declare-change response on or after the change on a genuine change trial; a *false alarm* is a declare response on a no-change trial; the hit rate H and false-alarm rate F over a trial batch yield sensitivity $d' = \Phi^{-1}(H) - \Phi^{-1}(F)$ and criterion $c = -\frac{1}{2}[\Phi^{-1}(H) + \Phi^{-1}(F)]$, with Φ^{-1} the standard-normal quantile function and rates clipped away from 0 and 1 before inversion. The central signal-detection analysis pairs this read-out with the graded attention clamp (Section 10.4.1): d' and c are computed as a function of the nominal clamp setting (with realized attention reported separately), isolating how attention moves sensitivity versus decision criterion (Section 4.3). In the Luo–Maunsell reward-structure sessions (Section 3) the same d'/c read-out is computed across the sensitivity and criterion sessions to test the sensitivity $\rightarrow d'$, criterion $\rightarrow c$ double dissociation [11]. Policy entropy (of the actor’s action distribution) and critic quantile spread are read out alongside d' and c under the clamp to quantify decision confidence and value uncertainty (Section 4.4).

10.4.3 Temporal decoding from working memory

To ask what the memory holds and when, a linear decoder is trained to read each trial-defining variable from the flattened working-memory state at each frame. On a batch of trials, cross-validated balanced accuracy is reported per frame $t = 1, \dots, 6$ for four variables: cue value (cue colour), cue validity (displayed ring proportion), change presence, and change location. Balanced accuracy is referenced against the chance level of each variable’s label set. This yields a per-variable time course across the trial — for example, whether a variable is latched at the cue and maintained, perceived transiently and then discarded, or constructed online at the change frame — and is the read-out behind the memory-code analysis of Section 4.5. The same decoder, applied to the value-directed versus pure-validity environments, estimates whether a variable remains linearly recoverable at the sampled frames. Manuscript claims about change location are restricted to the supported VDA4 analysis: the 2026-07-07 legacy `decode.npz` VDA2 and VDA9 change-location fields were generated with degenerate labels and are not interpreted here. Other fields in that archive, including the validity time courses reported in Section 6.5, are separate decoder targets and remain usable. No cue-location decoder was fitted across the blank delay; later cue-directed attention is reported as re-acquisition compatible with maintenance, not as a decoded maintained cue-location representation.

For VDA4 validity, the canonical manuscript series is the validity field for the affine-feedback VDA4 condition in the preserved `decode.npz` archive. It was generated by the set-size decoder harness on the canonical 128-memory VDA4 checkpoint; the evidence ledger records the full artifact paths. The series is 0.773 at $t = 1$, 0.296 at $t = 2$, and near four-way chance thereafter (mean $t = 2-6$: 0.309). A distinct exploratory deep-dive archive used a different sampling harness but does not independently encode sufficient checkpoint provenance for its validity series, so it is not used in the manuscript. The versioned corrected decoder collector writes beneath the dated derived-output directory; a completed corrected artifact remains pending, while the preserved legacy validity fields remain usable.

10.5 Reproducibility and implementation

Each reported analysis identifies a frozen set of weights, and reported condition means are estimated over freshly sampled trial batches at the operating point stated with each result. The set-size summaries use one canonical affine-feedback checkpoint per condition; the current figures do not provide between-seed uncertainty or saved trial-level confidence intervals. The manuscript does not use the legacy high-token `crossattn1` clamp outputs, whose location indexing failed to bias both the image and corresponding memory key at VDA9; all reported set-size clamp curves are for affine feedback. The environments, network, training loop, and analysis harnesses are implemented in Python with PyTorch. Training and analysis were run on Apple-silicon (MPS) and CPU backends; results are backend-independent, and the analyses that generate the reported numbers are run on the frozen trained network.

A Supplementary material

Figure inventory. The analyses of Section 4 are rendered from the frozen analysis outputs of the trained model: psychometric and reaction-time curves, per-condition attention super-figures and the α_c time-course, the signal-detection $c(\alpha_c)$ and $d'(\alpha_c)$ curves under the graded attention clamp, the distributional-critic uncertainty panels, the temporal-decoding curves, and the microstimulation false-detection curves. The value-directed results (Section 5) and the preliminary Luo–Maunsell comparison (Section 3) are rendered from their respective analysis harnesses. Figure sources are noted inline in each section as % FIGURE: comments.

Current battery status. The VDA1/2/4/9 affine $d_{\text{mem}} = 128$ battery is reported in Section 6; it is a completed analysis in this manuscript rather than a promised future appendix. Each point uses one canonical checkpoint with a 20,000-row logged phase, and VDA2 uses the `_2x2` checkpoint. VDA16 remains incomplete at checkpoint 599 and is not a quantitative endpoint. Preserved $d_{\text{mem}} = 256$ checkpoints exist, but the matched width battery is pending (Section 7). The delay-probe environment is implemented, but no delay-probe or causal delay-clamp result is reported (Section 8).

Invalid legacy artifacts. The 2026-07-07 NPZ archives are retained for provenance rather than overwritten. The VDA2/VDA9 change-location fields in `vda_sweep/figs/decode.npz` are invalid because the legacy collector produced degenerate labels. The high-token VDA9 `crossattn1` clamp fields in `vda_sweep/figs/sdt.npz` are also invalid for cross-routing comparison because the legacy indexer failed to bias the paired memory key. Neither field supports a claim in this affine-only manuscript; the evidence ledger records the field-level status without altering the archives. Corrected artifacts are pending in a versioned derived path.

Failed filtering-task trace. The attend-here/ignore-here training failure described in the Discussion is recorded in the preserved run summary `./ACT_II_RESULTS.md`, lines 68–77: both attempted checkpoints remained at the starting difficulty and collapsed to always waiting (zero declarations on both cued and foil-change trials). This trace is a failure record, not a quantitative endpoint.

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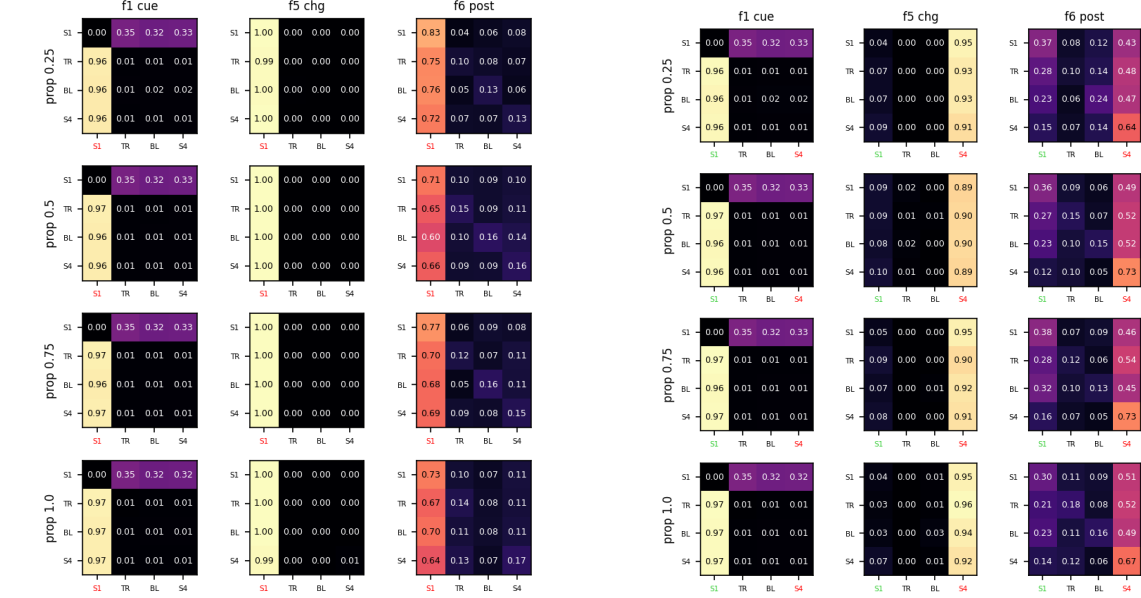
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affine_ew self-attn A[query→key] — valid change, cue@S1 (green=cued, red=changed) iter 4199

affine_ew self-attn A[query→key] — invalid change, cue@S1 (green=cued, red=changed) iter 8199



affine_ew A[query→key] by cue value (cue@S1 valid) iter 4199

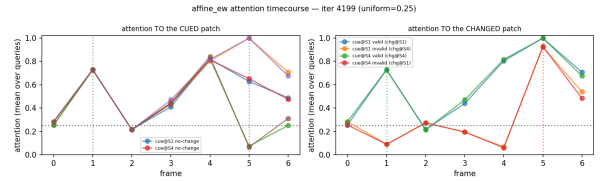
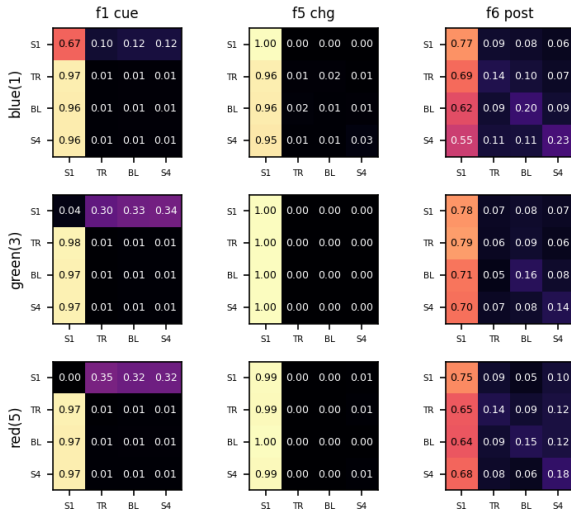


Figure 6: **Attention maps of the model, conditioned on cue validity and value.** *Top:* the per-patch attention map across the trial on validly cued (left) and invalidly cued (right) trials, resolved by displayed cue proportion — attention orients to the cue and then locks onto the changed patch. *Bottom left:* the attention maps split by cue colour (value) are essentially indistinguishable — value is *not* written into the priority map. *Bottom right:* the α_c time-course underlying the maps (orient; release at the blank; re-acquire; lock). All panels are the model (affine feedback) on *vda4*.

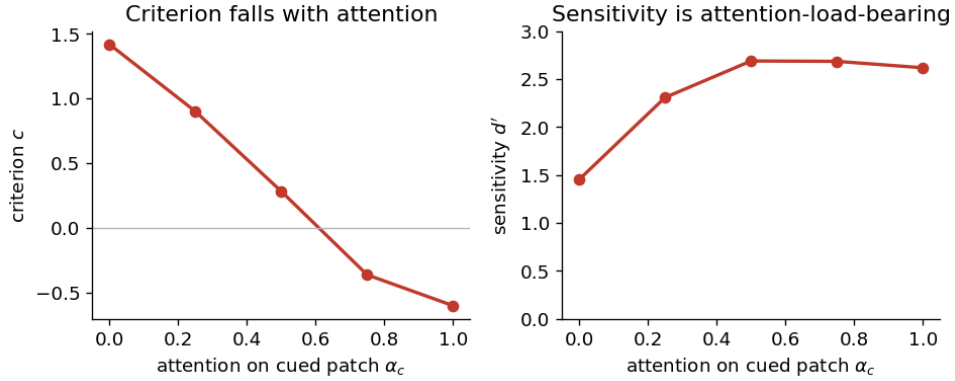


Figure 7: Signal-detection read-out under the graded cued-patch attention clamp (`vda4`, the model). **Left:** at these sampled VDA4 settings, the criterion c falls monotonically with attention, from 1.42 (suppress) to -0.60 (enhance) — attention makes the detector more liberal. **Right:** sensitivity d' is attention-mediated, collapsing to 0.63–1.45 under suppression and recovering to ~ 2.0 –2.7 under natural/enhanced attention. Both the criterion and the sensitivity account are realized by one manipulation.

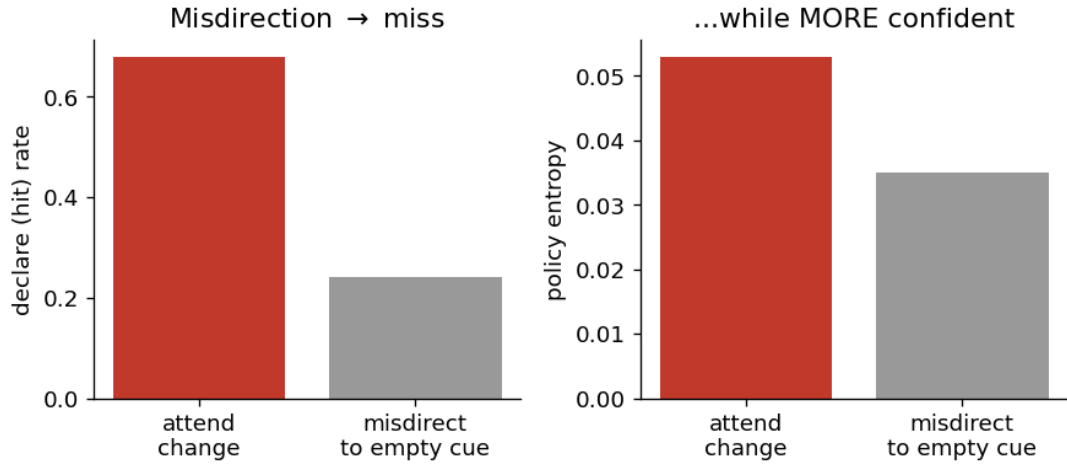


Figure 8: Decision uncertainty under attention misdirection (real change at the uncued patch, the model). Misdirecting attention onto the empty cued patch drops the declare rate from 0.68 to 0.24 (a miss) while lowering policy entropy from 0.053 to 0.035 (increased confidence) — a confident miss. The critic’s quantile spread widens under misdirection, so the value system registers the problem the policy commits to.

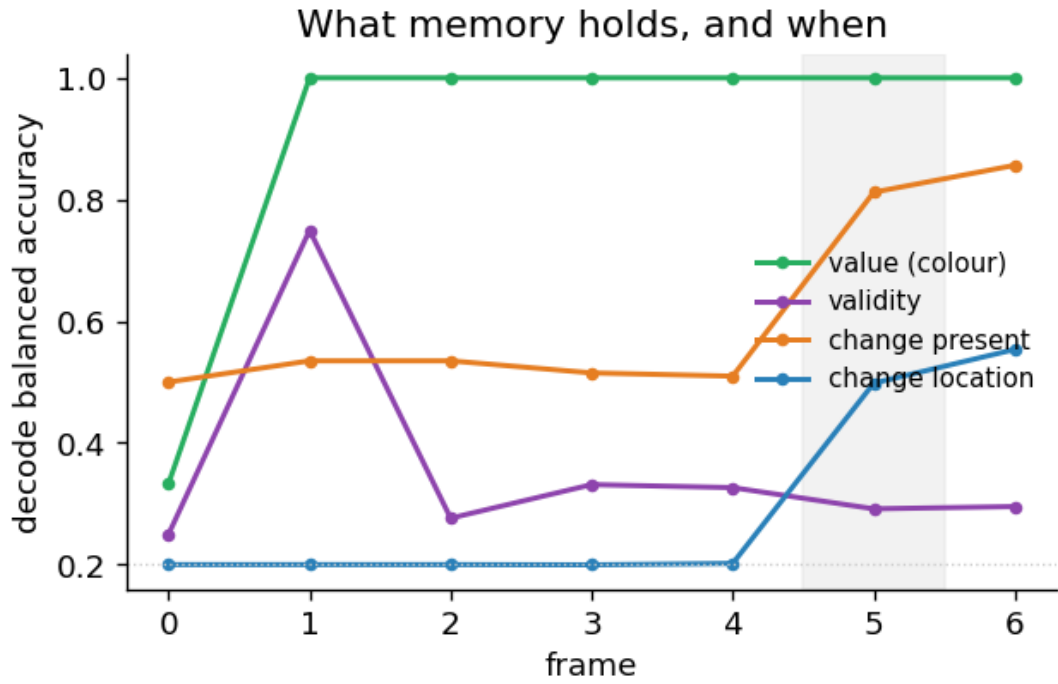


Figure 9: Temporal decoding of task variables from the recurrent working-memory state (the model). Value (cue colour) is latched at the cue and held at balanced accuracy 1.00. Validity (ring proportion) peaks at the cue (0.773 at $t = 1$), falls near chance at the blank (0.296 at $t = 2$), and remains weak thereafter (mean $t = 2-6$: 0.309). Change presence (0.81 at $t = 5$) and change location (0.50 at $t = 5$) are flat at chance until the change frame, then jump: the change is built online. The position-specific post-change code is compatible with a parietal-like event-location signal; no cue-location decoder was fitted across the delay.

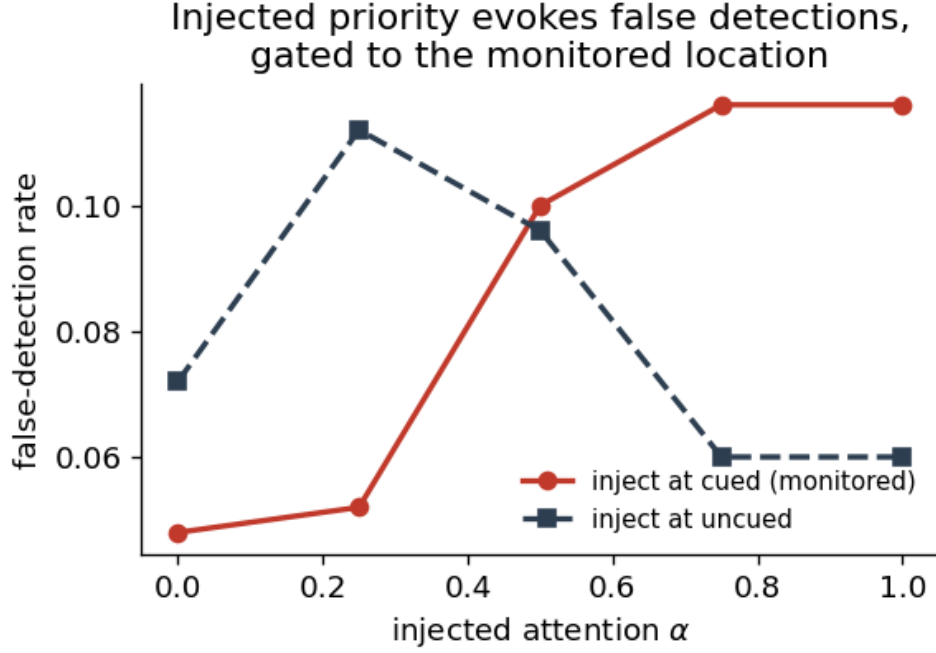


Figure 10: Attentional microstimulation on no-change trials (the model). Injecting attention at the cued (monitored) patch drives the false-detection rate from ≈ 0.03 to ≈ 0.12 (a $\sim 4\times$ increase); injecting the same attention at an uncued patch leaves it flat (≈ 0.05 – 0.10). The evoked false percept is spatially gated to the location the model is already monitoring.

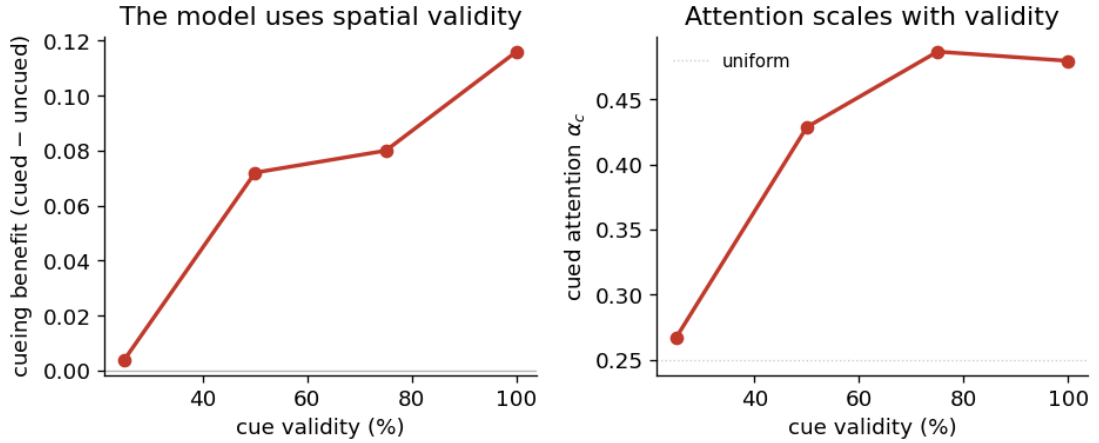


Figure 11: **A separately trained Validity4 checkpoint uses spatial validity.** With no competing value cue, the cueing benefit (cued – uncued detection at $\Delta = 26^\circ$; left) grows monotonically with cue validity, and cued-patch attention α_c (right) rises from near the uniform baseline (0.25) toward 0.48 as the ring signals higher reliability. Curves are the model (affine feedback).

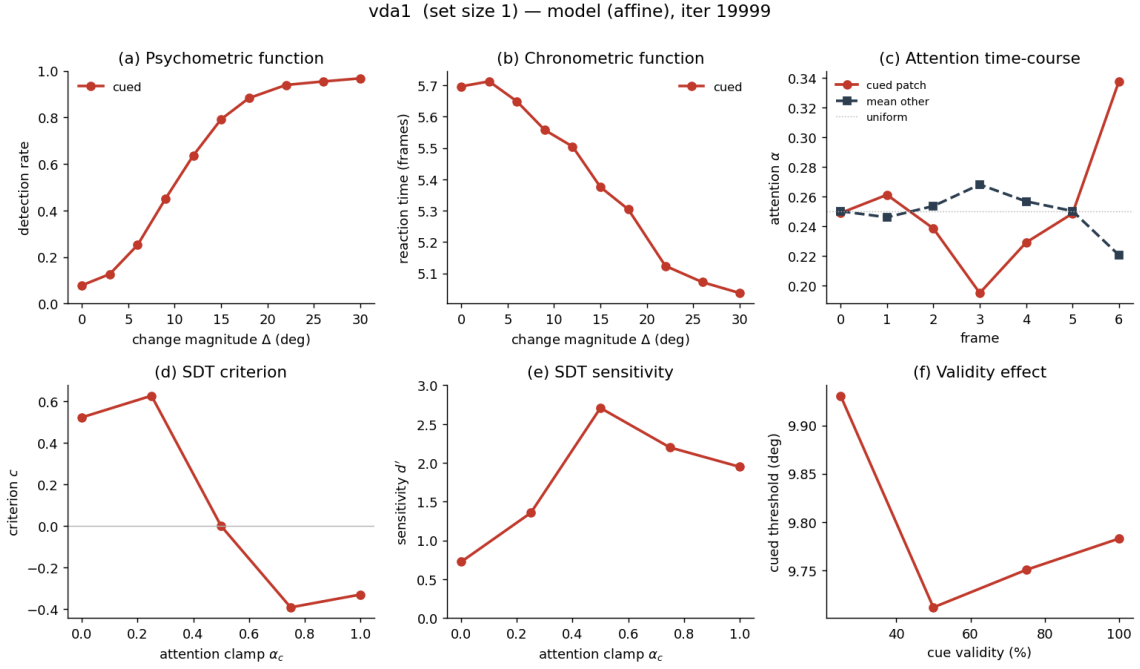


Figure 12: **Set size 1 (vda1): a single monitored stimulus.** With no spatial competition there is no cued/uncued distinction, so the psychometric and chronometric functions are cued-only. The mean cued detection threshold is $\approx 9.8^\circ$, the lowest of the battery — detection is easiest when the target is alone. The attention clamp produces the *largest* sensitivity modulation of any set size (d' swept from 0.72 to 2.71), i.e. at set size 1 attention is almost entirely a sensitivity (gain) effect; the criterion also moves ($c : 0.52 \rightarrow -0.33$).

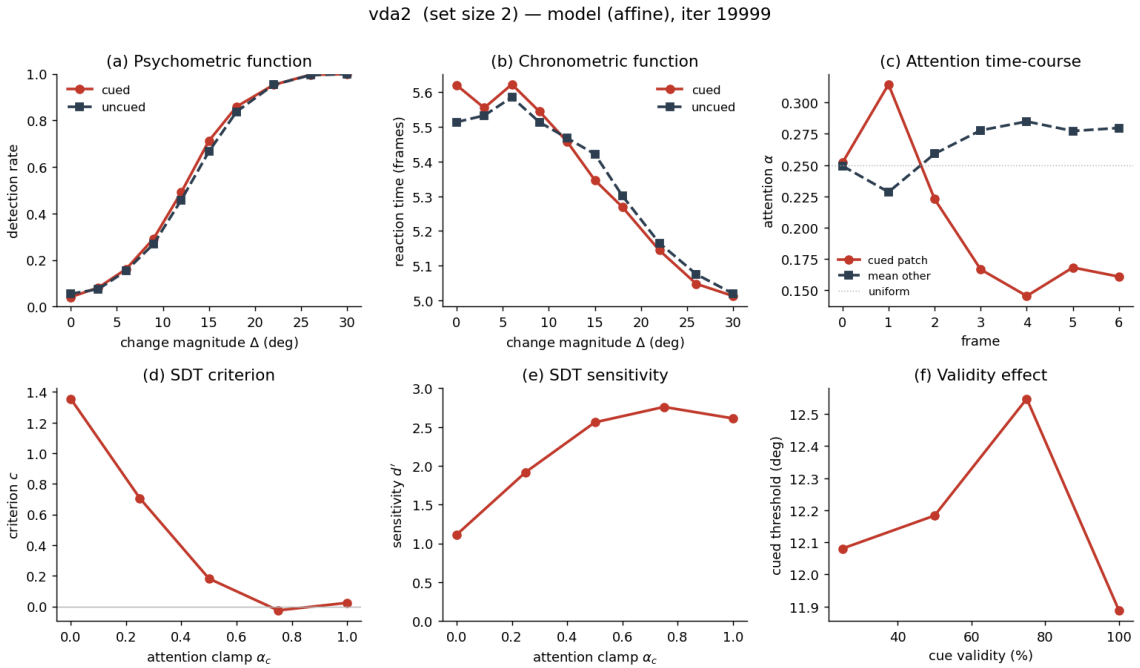


Figure 13: **Set size 2 (canonical vda2_2x2 checkpoint).** In this separately trained task, the cued threshold is $\approx 12.2^\circ$ and panel (a) shows a cued-over-uncued benefit. This cross-task contrast does not isolate set size. The criterion sweep is the widest of the battery ($c : 1.36 \rightarrow 0.02$), while the sensitivity modulation begins to compress ($d' : 1.11\text{--}2.76$).

vda4 (set size 4) — model (affine), iter 19999

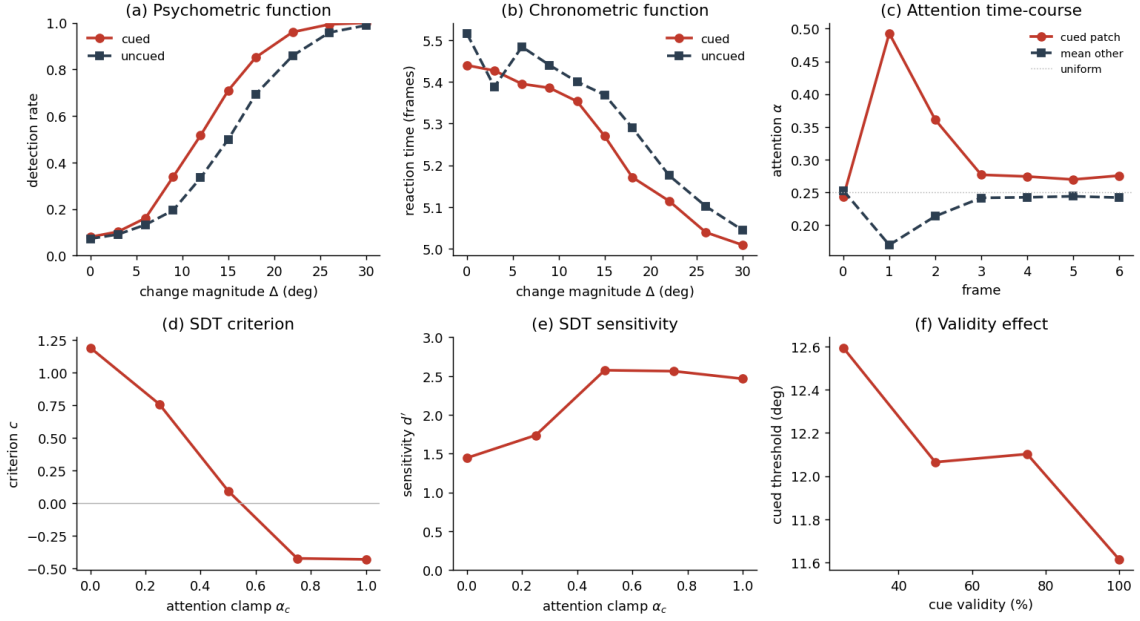


Figure 14: **Set size 4 (vda4): the canonical published condition.** The cued threshold is $\approx 12.1^\circ$; attention lowers the criterion strongly ($c : 1.19 \rightarrow -0.43$) and still moves sensitivity ($d' : 1.44\text{--}2.57$). The attention time-course (panel c) shows the model orienting to the cue and locking onto the changed patch, and the map is validity- and value-invariant (Section 4).

vda9 (set size 9) — model (affine), iter 19999

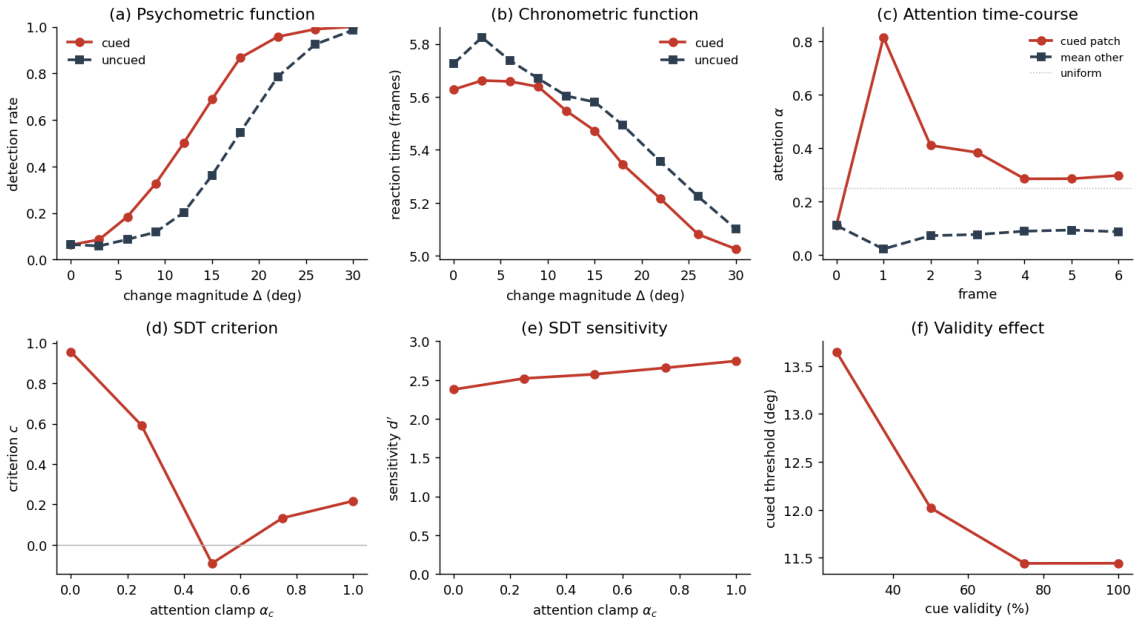


Figure 15: **Set size 9 (vda9): a 3×3 array.** The validity effect is now pronounced (panel f: the cued threshold falls steeply as the cue becomes more reliable). The nominal attention clamp has the smallest observed sensitivity range in the battery ($d' : 2.38\text{--}2.74$). This endpoint range alone does not establish saturation or show that sensitivity has ceased to contribute; the criterion range is also narrower than at set size 4 ($c : 0.96 \rightarrow 0.22$), with a high-clamp reversal visible in the full curve.

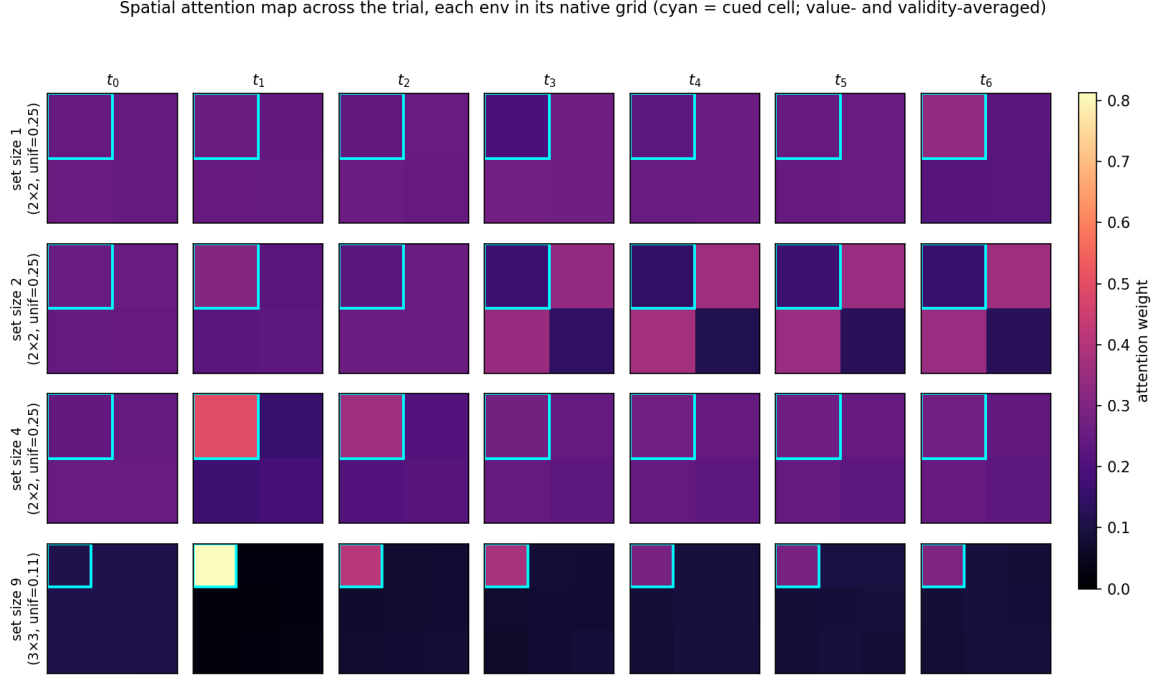


Figure 16: **The spatial attention map across the trial, per set size.** Each row is one environment; each column is one of the seven trial frames; each panel is the attention map in that environment’s *native* grid — 2×2 for set sizes 1–4 and 3×3 for set size 9 — with the cued cell (top-left) outlined in cyan (value- and validity-averaged; common colour scale; per-row uniform baseline noted at left). Attention concentrates on the cued cell at the cue frame t_1 and relaxes thereafter; cue-time concentration is largest at set size 9, whereas sustained elevation is not strictly monotone across all four checkpoints. At VDA9, $t_1 \approx 0.82$ against a uniform baseline of 0.11 for the nine-cell array, with ≈ 0.02 on each other cell. Separate affine-feedback checkpoints; VDA9 also differs in grid and token/readout dimensions.

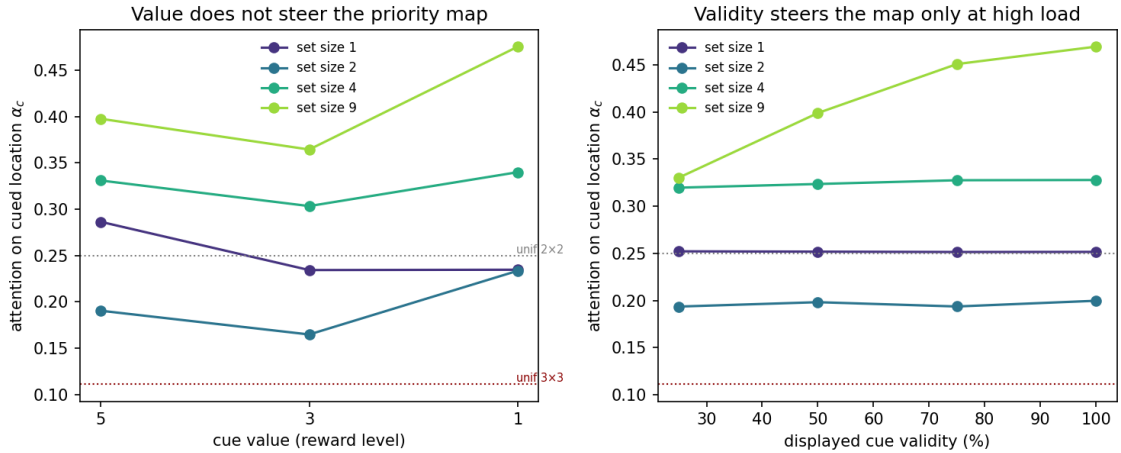


Figure 17: **Priority-map associations with value and validity differ across checkpoints.** *Left:* cued-location attention α_c as a function of cue value (reward level 5/3/1), one curve per set size — small and non-monotone at every set size (no value steering). *Right:* α_c as a function of displayed cue validity — nearly flat in VDA1/2/4 and increasing in VDA9. Uniform baselines are task-specific: 0.25 for the 2×2 VDA1/2/4 tasks and $1/9 \approx 0.11$ for VDA9. Separate affine-feedback checkpoints; the VDA9 contrast is associated with, but does not isolate, load.

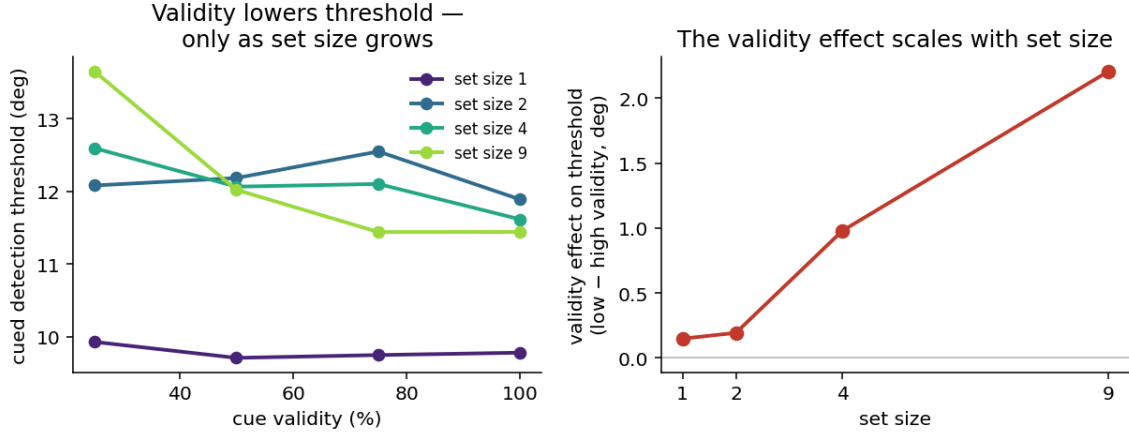


Figure 18: **Low-to-high-validity cued-threshold changes across checkpoints.** *Left:* cued detection threshold vs. displayed cue validity, one curve per set size — flat at VDA1, steeply decreasing in VDA9. *Right:* the displayed-validity effect is defined as low-validity minus high-validity *cued* threshold and is 0.148° , 0.193° , 0.979° , and 2.207° ; it is not a cued-versus-uncued benefit. Separate affine-feedback checkpoints.

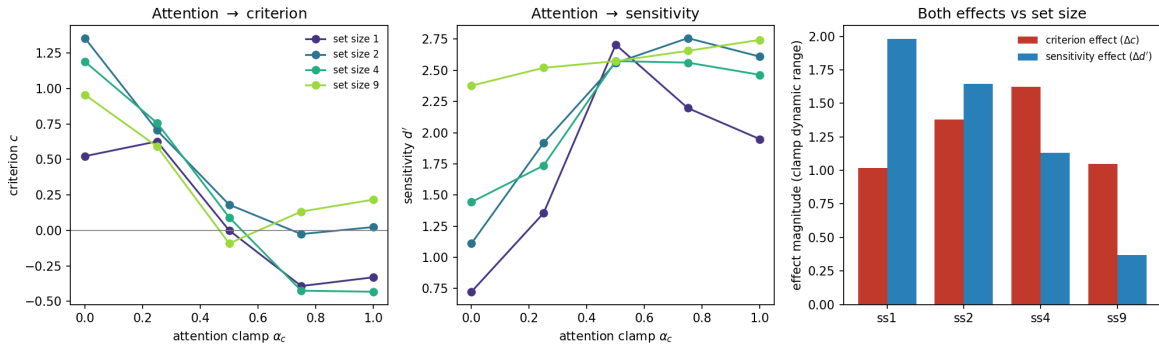


Figure 19: **Criterion and sensitivity across set size.** *Left:* criterion c vs. the attention clamp, one curve per set size. *Middle:* sensitivity d' vs. the clamp. *Right:* the magnitude of each effect (clamp range) against set size. Sensitivity ranges are 1.982, 1.647, 1.131, and 0.367; criterion ranges are 1.020, 1.381, 1.622, and 1.048. These are clamp ranges, not the displayed-validity threshold improvements. VDA9 criterion is not uniformly monotone.

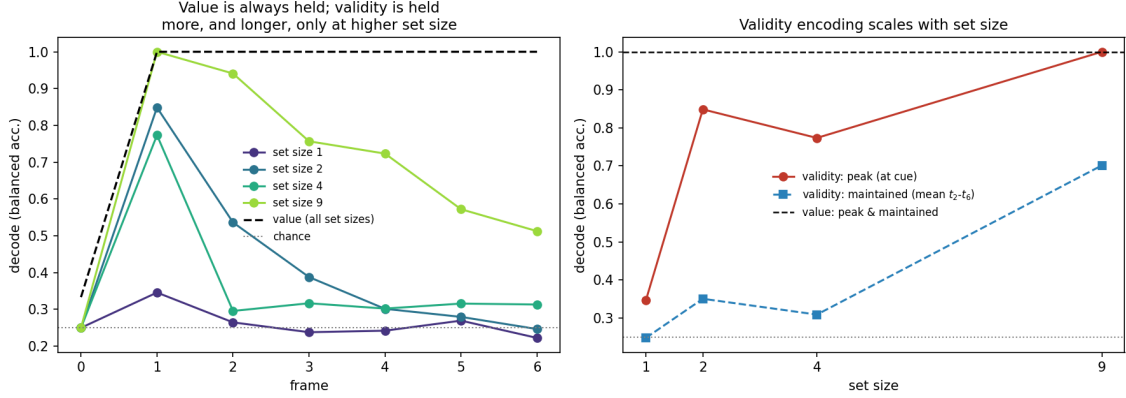


Figure 20: **Validity-decoding profiles differ across checkpoints, while value remains recoverable at ceiling.** *Left:* balanced accuracy of decoding displayed cue validity from the recurrent memory state, per frame, one solid curve per set size (chance 0.25); the dashed black line is the value (cue-colour) decode, pinned at ≈ 1.0 across all set sizes. Validity decays after the cue except at set size 9; the value decode remains near ceiling. *Right:* the peak (at-cue) and maintained (mean over post-cue frames) validity decoding is highest at VDA9 but not monotone at the intermediate checkpoints, whereas value (dashed) is at ceiling for peak and maintained at every set size. Separate affine-feedback checkpoints.

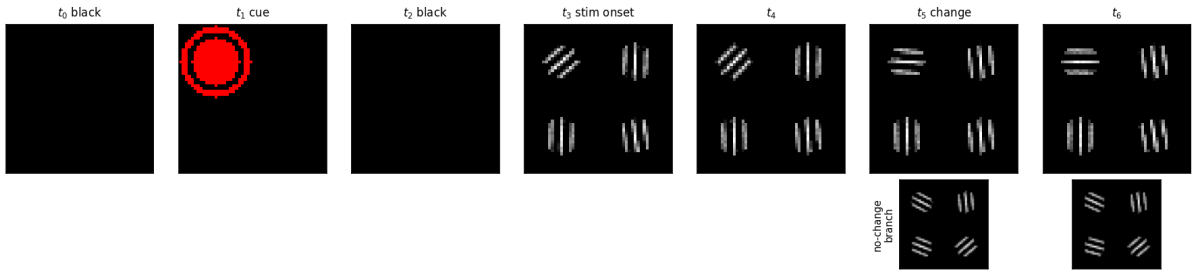


Figure 21: **Trial timeline of the change-detection environment.** Each trial is seven frames: a blank, the cue (whose ring completeness signals validity and whose colour signals value), a blank delay, and four held Gabor patches, one of which steps in orientation by Δ at the fixed change frame $t = 5$ on half of trials. The model chooses *wait* or *declare-change* at every frame.